

Detecting Fraud Signals in Centers for Medicaid and Medicare Services (CMS) Data using Machine Learning on Class Imbalanced Big Data

Sabrina Woo, Biohealth Analytics

Narsim Kamath, High Performance Computing

Cameron Kass, High Performance Computing

Love Ayinde, High Performance Computing

DSA Mentor: Dr. Timothy Haithcoat

Subject Matter Experts:

Director Arvids Petersons of the Missouri Medicaid Fraud Control Unit; Missouri Medicaid Audit and Compliance team, including Director Richard Ferrari, Darla Weekley, Colin Murdick and Jeffrey Rex; Lucas Gallardo from Alivia Analytics.

Abstract

Fraud is a pervasive and complex problem within the United States healthcare system necessitating evermore complex methods of detection to uncover increasingly convoluted schemes. Our approach uses an analytical framework that integrates Centers for Medicare & Medicaid Services (CMS) Durable Medical Equipment, Prosthetic Devices, Prosthetics, Orthotics, & Supplies (DMEPOS) claims, NPPES, Open Payments, the Office of Inspector General (OIG) Exclusion List, and U.S. Census indicators into a standardized provider-year dataset. We apply an explainable unsupervised ensemble to model baseline “normal” behavior within each specialty, then flag providers who are statistically atypical without relying on large numbers of confirmed fraud labels. Key signals include year-over-year changes, z-scored billing and payment spikes, and utilization mix patterns derived from HCPCS codes and product categories. Each provider receives a risk score by scaling and aggregating anomaly outputs across models (K-Means, LOF, Isolation Forest), producing a ranked list for triage. For transparency, the score is paired with top contributing anomaly factors and an evidence summary that documents the specific behaviors driving the flag, enabling faster, more consistent investigative prioritization. We determine that the top HCPCS service codes billed by fraudulent referring providers demonstrate a significant association with fraud compared to their prevalence in the general population of referring providers for the Medicare DMEPOS program over the period studied. We separately conclude that supervised machine learning methodology is not amenable to the extreme class imbalanced scenario of Medicare fraud.

Audience, Project Definition, Background, and Aims or Objective

Intended Audience

Our project is designed for a broad network of fraud prevention stakeholders, including CMS Medicare fraud teams, the Office of Inspector General (OIG), State Medicaid Fraud Control Units (MFCUs), and state program integrity units. The CMS Medicare Fraud Strike Force, jointly led by the Department of Justice (DOJ) and the Department of Health and Human Services (HHS) OIG, targets large-scale Medicare fraud schemes. Supporting these enforcement efforts, state program integrity units within Medicaid agencies focus on preventing improper payments, conducting audits, and referring suspicious cases for further review. State Attorneys General (AGs), and MFCUs investigate and prosecute fraud, waste, and abuse in federally funded healthcare programs through both criminal and civil actions. Collectively, these entities form a unified system dedicated to protecting public healthcare funds and preserving the integrity of Medicare and Medicaid programs.

The impact of our project has both direct and indirect consequences. Preventing FWA in socially funded programs such as Medicare and Medicaid provides an indirect financial benefit to all U.S. taxpayers by preserving public funds and ensuring better financial sustainability of these programs. In addition, this project offers a direct business benefit for fraud investigation teams by generating leads on potential fraud schemes and possibly uncovering national trends. Collectively, these outcomes reduce the systemic risk of undetected FWA and strengthen institutional accountability.

Background

Healthcare fraud, waste, and abuse (FWA) drain billions of taxpayer dollars each year and undermine the integrity of federally funded programs, posing a persistent challenge to the U.S. healthcare system. In fact, FY2024 estimates indicate that improper payments totaled \$31.7 billion in Medicare Fee-for-Service and \$31.1 billion in Medicaid ([CMS, Fiscal Year 2024 Improper Payments Fact Sheet](#)). One particularly vulnerable area involves Durable Medical Equipment, Prosthetics, Orthotics, and Supplies (DMEPOS), which includes items such as wheelchairs, knee braces, and other medical supplies. According to the Office of Inspector General (OIG), Medicare payments for DMEPOS exceed \$7 billion annually, and this category remains “a major concern” for FWA in federal programs ([OIG, Durable Medical Equipment Fraud and Safeguards in Medicare](#)).

The U.S. legal and enforcement landscape has increasingly prioritized combating FWA, establishing it as a strategic focus of federal healthcare oversight. For example, on June 30, 2025, the U.S. Department of Justice announced the largest healthcare fraud takedown in U.S. history, charging 324 defendants in schemes amounting to \$14.6 billion ([United States Department of Justice, National Health Care Fraud Takedown](#)). Although there has been a revitalized focus, FWA investigations rely heavily on public hotline referrals, interagency recommendations, manual audits, and retrospective reviews, which are often slow, resource-intensive, and reactive rather than preventive. Our project will explore how data-driven and predictive approaches can identify providers whose billing and financial behaviors deviate from expected norms. Using a machine-learning framework to analyze provider-level claims data and socioeconomic data to predict potential DME-related fraud, we will be positioning the integrity and anti-fraud agencies to stay ahead of emerging AI driven schemes.

Literature Review

Similar Work

Du Preez, A., Bhattacharya, S., Beling, P., & Bowen, E. (2025). This study conducts a systematic literature review of machine learning approaches used in health insurance fraud detection, examining data sources and methodologies reported over the past 20 years. The review highlights key challenges in the field and identifies opportunities for future research. This paper is one of the most important papers for our project, given its comprehensive and thorough literature review of available datasets and methodology. The paper dives into the uses of supervised and unsupervised learning in our subject space and details the most common features and practices for working with healthcare data. It also gives a large list of example tools and packages available that have been used in other studies.

Li et. al (2008). The paper provides a comprehensive survey of statistical methods for healthcare fraud detection, examining both supervised and unsupervised approaches to identifying fraudulent behaviors in medical data. The paper categorizes supervised methods, which require labeled datasets of known fraud cases, including techniques such as neural networks, decision trees, and Bayesian networks, as well as unsupervised methods, which detect anomalies or outliers without pre-labeled data, using clustering and anomaly detection algorithms. It also emphasizes the importance of data preprocessing, including cleaning, normalization, and feature selection, to improve the accuracy and reliability of fraud detection models. By reviewing the strengths, limitations, and applications of various statistical techniques, the paper offers critical insights for healthcare fraud analytics, highlighting how combining different approaches can enhance detection performance. This survey is directly relevant to the project on identifying healthcare provider fraud, as it provides foundational knowledge for selecting appropriate methods, integrating provider-level and socioeconomic data, and ensuring robust preprocessing to build predictive models capable of scoring and ranking high-risk providers.

Bauder and Khoshgoftaar are popular publishers in this area, having conducted multiple machine-learning studies using Medicare data for fraud. *Johnson, J. M., & Khoshgoftaar, T. M. (2023)* leverages Centers for Medicare & Medicaid Services (CMS) data for Medicare Part B, Part D, and Durable Medical Equipment, Prosthetics, Orthotics, and Supplies (DMEPOS) programs during the 2013 – 2019 program years to perform fraud classification with extreme gradient boosting (XGB) and random forest (RF). Additionally, it details a methodology for labeling records as fraudulent based on a subset of exclusion rules applied to List of Excluded Individuals and Entities (LEIE) datasets for each year of Medicare data. It further provides a blueprint for engineering features from the Medicare programs' datasets that we may follow to validate whether predictors of fraud have changed over time. We will replicate the labeling procedures from this article in our project and feature engineering steps on numerical features. Another such example *Bauder, R. A., & Khoshgoftaar, T. M. (2018)* uses random forest models while investigating how class imbalance in Medicare big data affects fraud detection. They apply random under sampling to create seven different class distributions (From 99.9:0.1 to 50:50) and test

random forest performance. The result showed that the default 50:50 balanced distribution is not optimal for fraud detection but the distribution of 90:10(non-fraud: fraud cases) is achieving the best AUC score. It was also noted that the 99.9:0.1 distribution has a reduced accuracy level. For our project we could carefully tune non-fraud/fraud ratios instead of defaulting to balanced datasets. By using under sampling or oversampling strategies on CMS billing and socioeconomic data, we could improve the model accuracy and avoid overfitting, ensuring our fraud risk scores reflect the most reliable detection patterns.

Tajrobehkar, M., Guo, X., Nguyen, D., Chigullapally, N., Shravah, V., Yee, S., Chozhan, A., & Hoang, K. (2025) explores the use of machine learning to improve fraud detection (specifically overutilization). It criticizes traditional methods that rely heavily on LEIE datasets and manual audits, arguing that the approach tends to ignore subtle patterns and fixates only on extreme cases. To address this issue, the author created a labeled dataset in ophthalmology with the help of medical billers and physicians. The dataset included features that were designed to capture realistic overutilization patterns. Multiple machine learning models were used for testing, such as K-nearest neighbor (KNN), logistic regression (LR), support vector machines (SVM), extreme gradient boosting (XGB), and multilayer perceptron (MLP). The study emphasizes that current CMS methods are too simplistic and ineffective when compared to machine learning, which could provide real-time detection and cost savings. However, it too has its own limitations; datasets are quite small, the predictions are not perfect, distinguishing legitimate overutilization from fraud is difficult, and scaling to other specialties would require labor-intensive expert-labeled datasets. For our project, this paper is quite useful, as it demonstrates how well-designed labeled datasets can enhance fraud detection using machine learning, without ever relying on the LEIE database. The issue is that if we were to apply this approach, we would be doing so on a broader scale with multiple specialties and different datasets, making it a bit harder to capture “subtle” patterns and possibly reducing accuracy, highlighting the challenges of scaling this methodology.

Bairy, M., Muniyal, B., & Shetty, N. P. (2024): This paper focused on detecting healthcare fraud using unsupervised learning methods that don’t require labeled fraud data. They apply techniques such as Isolation Forest, Local Outlier Factor (LOF), kNN-based outlier detection, and autoencoders to identify anomalous billing and provider behaviors, comparing which approaches most effectively flag suspicious patterns. For our project, this is highly relevant because we also face limited ground truth labels; integrating these unsupervised methods can help us surface abnormal providers or claims as high-risk cases. These anomaly scores could then feed into our fraud risk scoring system and support our dashboard visualizations, strengthening fraud detection even when labeled data is sparse.

Kaddi, S. S., & Patil, M. M. (2023) This paper addresses the challenges of detecting fraud in healthcare claims within imbalanced datasets. Due to the evolving nature of fraudulent schemes, standard rule-based fraud detection methods are ineffective. Therefore, the authors proposed an ensemble learning approach to overcome this major hurdle. A stack ensemble learning model was specifically utilized in this paper, which combines multiple base classifiers to improve accuracy. Given the extremely small proportion of fraud cases, the Synthetic

Minority Over-sampling Technique (SMOTE) was employed to balance the dataset and improve performance. The performance of the model was compared with individual classifiers, and the experiment showed that the ensemble learning approach significantly outperforms the rest. For our project, this model could be extremely useful due to its strong predictive capabilities and could even serve as a reliable benchmark to compare different approaches, considering the characteristics of our dataset. Additionally, it provides methodological guidance on combining classifiers, which can further strengthen our modeling approach.

El-Lanky, M., & Chrimes, D. (2024). This paper presents a big data analytics approach to detecting fraud in U.S. Medicare claims, emphasizing how the massive scale of healthcare service data can be effectively processed to uncover irregular billing and provider behaviors. They address the challenges of big data by leveraging parallel computing and distributed frameworks such as Hadoop and Spark to handle millions of claims efficiently, while also reducing complexity through feature engineering that aggregates raw claims into provider- and service-level metrics. This not only makes the data computationally manageable but also highlights patterns that are meaningful for fraud detection across large and diverse populations. Their work demonstrates the importance of scalability, ensuring that fraud detection methods maintain performance when applied to nationwide Medicare claims. For our project, their approach provides a roadmap for managing CMS billing data at scale: by adopting big data platforms and aggregating claims into structured, provider-level features, we can ensure our fraud risk scoring system remains both efficient and effective, allowing us to capture meaningful fraud signals without being overwhelmed by the raw size and complexity of the datasets.

Hamid, Z., Khalique, F., Mahmood, S., Daud, A., Bukhari, A., & Alshemaimri, B. (2024). This paper addresses the challenge of healthcare insurance fraud detection by designing a hybrid two-stage pipeline that combines association rule mining with unsupervised anomaly detection. Using the Apriori algorithm, they first extract frequent patterns in claims, then apply methods such as Isolation Forest, CBLO, ECOD, and One-Class SVM to flag usual behaviors. Using the Silhouette score, the methods that perform best are CBLOF and Isolation Forest. This approach highlights how reducing the data into transaction rules before anomaly detection improves both efficiency and accuracy. For our project, their frameworks could be a road map we could exploit using the more recent year data and/ or the CMS billing data. I believe the anomaly detection could be used to uncover outlier providers, and flagged anomalies could be part of the variables that influence our fraud risk scoring system and should be integrated into our dashboard and help investigators identify high-risk providers in Missouri more effectively.

Hancock, J.T., Bauder, R.A., Wang, H. et al. (2023) developed a machine learning framework for Medicare fraud detection. They introduced a novel ensemble supervised feature selection using random forest, decision tree, and gradient boosted decision trees to rank features and select the median rank across all models. They then performed ANOVA and HSD to find the minimum number of features for selection.

Liu, Q., & Vasarhelyi, M. (2013). This paper utilizes an unsupervised clustering model that incorporates geo-location data. Using insurance claims data, the models calculate the distance between the providers and the beneficiaries to identify unusual patterns, such as patients travelling large distances for service. It then clusters the providers and claims based on the geography and service patterns to highlight outliers without requiring labeled data. The author also emphasizes that combining unsupervised clustering with supervised models could enhance the predictive power. We could adopt a similar approach using Medicaid/Medicare Part B claims data and extend it by incorporating socioeconomic factors to improve accuracy in identifying regional patterns of risk. These insights can further enhance our deliverables as well.

Sethi, S. (2025) focuses on the broader concept of risk scoring in healthcare analytics, particularly how machine learning models can assign quantitative risk scores to entities such as patients, providers, and clinical processes to predict high-risk outcomes. The primary focus was on clinical risk, such as hospital readmission and compliance issues. The paper emphasized model interpretability, feature selection, and regulatory compliance as key considerations when implementing risk scoring systems in real-world healthcare settings. The score was determined by integrating multiple predictive features from historical and real-time data, which may include billing patterns, claim volumes, and frequency of high-cost services, unusual service combinations, prior sanctions or exclusions (like in the LEIE database), and regional socioeconomic indicators. Each feature is typically standardized or normalized to ensure comparability and may be weighted based on its predictive importance, which can be determined through machine learning techniques. The model then calculates an overall risk score for each provider, usually scaled between 0 and 1 or 0 and 100, where higher scores indicate a higher likelihood of fraudulent or anomalous behavior. Advanced implementations may also provide feature-level contributions, explaining which factors most influenced a provider's score, enhancing interpretability and enabling stakeholders to understand why a provider is flagged as high risk. In practice, these scores can be used to prioritize investigations, allocate audit resources efficiently, and feed visual dashboards that highlight high-risk providers and regions, while maintaining regulatory compliance and transparency. By combining predictive modeling with scoring, your project can quantify fraud likelihood systematically, compare providers across multiple dimensions, and produce actionable insights for CMS, OIG, and state fraud units.

Feature Engineering

Kumaraswamy, N., Markey, M.K., Barner, J.C., Rascati, K. (2022). This paper highlights and introduces a feature engineering framework that includes data aggregation of raw transactional data at a provider level, feature creation based on domain knowledge, followed by the use of statistics like mean, median, sum, and standard deviation, PCA to handle multicollinearity and dimension reduction, and utilization of ML models (Logistic and Random Forest) to perform validation. The 176 engineered features were reduced to 15 principal components that captured 85% of the variance in the data. With these features, the models displayed improved performance in identifying potential fraudulent providers, highlighting the value of careful feature engineering. Even though our project uses different, publicly available data, the systematic feature extraction, aggregation, and validation

documented within this paper can act as a guide in the development of effective models for our analysis. Additionally, the paper discusses its results while also comparing it with results from other studies that have used multi-source datasets for fraud detection, highlighting the possible benefits of incorporating data from different data sources.

Aljufri, M. A., Widyadhari, S., Asmani, A., Umam, A. M. and Purwitasari, D. (2025). This paper discusses the combination of health-system expertise with feature engineering and unsupervised models to identify hidden fraudulent patterns in national health-insurance data. Feature engineering played a crucial role. Instead of solely relying on raw administrative data, the authors constructed features based on their extensive domain knowledge by combining multiple BPJS health administration datasets, which were then grouped into major fraud schemas: upcoding, falsified diagnoses and medical records, and Non-existent billing, claims, and services. This type of feature mapping allowed clustering models to identify meaningful outliers that correspond to actual fraud behavior. Similar to our project, unlabeled data is being used; as such, the strategies mentioned in the paper could be incorporated into our work. Linking our datasets by NPI and creating fraud-relevant features related to costs/billing, frequency(utilization and overutilization), ownership, and even temporal patterns, along with the use of unsupervised models, could bear fruitful results.

Popular Machine Learning Methods

Hernandez et al. (2024): Some of the most frequently applied models in financial fraud research include (from least to most popular): long short-term memory (LSTM) networks, autoencoders, XGBoost, k-nearest neighbors, artificial neural networks, Naïve Bayes, support vector machines, decision trees, logistic regression, and random forests. Supervised learning remains the dominant approach in financial fraud studies. Common supervised learning techniques include LR, SVM, DT, RF, KNN, NB, and ANN. Unsupervised learning represents a quarter of references with methods including DBSCAN, autoencoders, and isolation forests. Finally, hybrid approaches that combine supervised, unsupervised, and semi-supervised learning account for very few mentions. These methods leverage both labeled and unlabeled data, proving especially useful when labeled data are limited or costly to obtain, as they aim to enhance model performance by utilizing additional information from unlabeled datasets.

Bolukbasi et al. (2025). This paper presents a comparative analysis of ten machine learning models which are: Random Forest, Support Vector Machines, Gradient Boosting, K-Nearest Neighbors, Naïve Bayes, Extreme Gradient Boosting, Voting Classifier, AdaBoost, Decision Tree, and Logistic Regression—for detecting insurance fraud. Among these, the Voting Classifier achieved the highest accuracy and F1-score of 87%, outperforming individual models through ensemble learning. To address class imbalance in the dataset, the authors implemented the Synthetic Minority Over-sampling Technique (SMOTE), which significantly improved model performance and robustness across all algorithms. A unique contribution of the study is its integration of Explainable Artificial Intelligence (XAI) methods, namely Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), to interpret the decisions made by complex models. Through these tools, the study

identified *incident severity* and *umbrella limit* as the most influential predictors of fraudulent behavior, while also confirming consistency between different interpretability techniques. By demonstrating how LIME and SHAP can jointly enhance model transparency and stakeholder trust, this research offers an important methodological bridge between predictive performance and explainability key aspects for fraud analytics in regulated domains such as healthcare and insurance.

Hybrid Models

George et al. (2025) identified literature that hybrid models that combine supervised, unsupervised, and deep learning strategies demonstrated superior adaptability and detection accuracy. *Carcillo et. Al (2021)* employs fraud detection using a hybrid model. This study presents a hybrid credit card fraud detection method that combines supervised and unsupervised learning to improve accuracy. The research compares outlier scores at different cluster levels and finds that cluster-based scores perform better than global or local ones. Results show that this combined approach enhances detection performance, particularly in terms of AUC-PR, although using too many outlier scores at once can reduce overall accuracy.

Logistic Regression

Suleiman et al. (2019) utilize Firth's penalized logistic regression to predict incorrect Diagnosis-Related Group (DRG) assignments in healthcare auditing, a context marked by rare events and pronounced separation across high-cardinality categorical predictors. In such settings, standard logistic regression often fails to converge when categories exhibit error rates of 0% or 100%, whereas Firth's approach retains these predictors through bias-reduced estimation. As a result, the authors obtain stable, finite coefficient estimates and superior out-of-sample performance, with the Firth-based model achieving the highest AUC among competing logistic and Bayesian approaches.

Similarly, *Iveniuk et al. (2025)* apply Firth's penalized logistic regression to model rare binary health outcomes in an aging population, where sparse predictors and imbalanced event rates undermine conventional logistic regression. The bias-reduced likelihood yields reliable parameter estimates and valid inference despite separation, underscoring the method's robustness and interpretability in healthcare applications involving rare outcomes. In relation to our project, this approach enables reliable Medicaid fraud risk estimation while preserving informative but sparsely observed predictors, thereby supporting accurate classification and effective audit prioritization in large-scale healthcare claims data.

Feature Selection

The determination of a feature selection methodology for is a significant challenge for our project due to the extreme class imbalance. We consider the approach proposed by *Matharaarachchi et. al (2021)* which orders features based on the sum of absolute values of principal component loadings combined with a sequential selection process. This method was shown to more effectively identify informative features, especially under various class imbalance levels and sample size conditions, outperforming traditional techniques like Recursive Feature Elimination using both synthetic simulations and real-world classification tasks.

SHapley Additive exPlanations (SHAP)

The authors *Lundberg and Lee (2017)* introduce SHapley Additive exPlanations (SHAP), a foundational framework for interpreting predictions from complex machine learning models using Shapley values from cooperative game theory. It shows that many popular explanation approaches can be expressed as additive feature attribution methods, where each feature receives a contribution score that sums to the model's output for a given instance. Within this class, the authors show that there is a unique scheme that satisfies three properties: local accuracy, missingness, and consistency. This scheme corresponds to Shapley values. SHAP is relevant because it makes fraud models more explainable to investigators, showing which inputs drive a specific fraud risk flag and support more transparent decisions. Other works mentioned in this literature review use it as a method to interpret the decisions of the machine learning models.

Unsupervised Learning

Breunig et al. (2000) is a foundational paper in the local outlier factor (LOF) method of anomaly detection. LOF is a density-based method for detecting objects that are unusual relative to their local neighborhood rather than across the global dataset. LOF assigns each object a continuous score by comparing the object's local density to the average local density of its nearest neighbors. Objects with substantially lower local density than their neighbors receive a high LOF score and are flagged as local outliers. Entities that operate in significantly sparser regions i.e., provider billing patterns that are unusual among their peers receive high LOF scores and are prioritized as potential fraud cases.

Evaluation Methods

Boyd, K., Eng, K. H., & Page, C. D. (2013). In this paper, the authors perform a computational analysis of estimators of area under the precision-recall curve (AUPRC) and their confidence intervals (CIs). Precision-recall curves, like receiver operating characteristic (ROC) curves, are evaluation tools for binary classification, and they have been increasingly used for imbalanced datasets like fraud. The authors find empirical evidence by testing on simulated data from known distributions favoring lower trapezoid, average precision, or interpolated median estimators for computing AUPRC, and they recommend binomial or logit CIs. They find that bootstrap and cross-validation (CV) estimators fail to produce valid 95% CIs, but they approach 95% coverage asymptotically as data size increases.

Problem / Question / Hypothesis

- Can interpretable fraud risk signals derived from publicly available transparency data, including Open Payments and ownership disclosures, identify elevated fraud risk among Durable Medical Equipment, Prosthetics, Orthotics, and Supplies (DMEPOS) providers?
- Which financial relationships, ownership interests, provider characteristics, geographic context, and temporal billing patterns are most strongly associated with elevated fraud risk at the specialty level?
- Can these interpretable signals be integrated into an explainable, specialty-level fraud risk score suitable for operational triage and investigation prioritization?
- How do ownership-based financial relationships differ across DMEPOS specialties, and are higher ownership concentrations associated with elevated fraud risk signals?
- Can interpretable feature attribution methods reveal consistent drivers of fraud risk across DMEPOS specialties, or do these risk factors vary substantially by specialty group?
- How effectively do specialty-level fraud risk signals prioritize investigation targets when evaluated against LEIE exclusion outcomes?
- How are specialty-level Medicaid fraud risk signals associated with socioeconomic conditions such as income, education, poverty, and insurance coverage measured using U.S. Census data?
- Do specialty-level fraud risk signals derived from integrated public datasets remain interpretable and stable across multiple feature groups, including payments, ownership, geography, and socioeconomic context?
- Can unsupervised anomaly detection methods identify high-risk DMEPOS specialties without relying on explicit fraud labels?
- How do specialty-level fraud risk signals differ when benchmarked against peer specialties rather than evaluated in isolation?
- Do specialty-level fraud risk signals capture transient provider behavior characterized by sudden entry, rapid increases in Medicaid billing activity, and subsequent sharp declines to near-zero utilization, consistent with short-lived or “pop-up” provider schemes?

Data Acquisition, Governance, Curation and Source Code Management

Description of Data and Data Sources

This case study utilizes only publicly available datasets released by U.S. federal agencies, all of which are described below. The data were accessed directly from the official government websites for each source. These datasets are intended for purposes of public transparency, research, and policy analysis. No restricted, proprietary, or patient-level identifiable data were used in this research.

Office of Inspector General (OIG) Exclusion List

The OIG Exclusion List is maintained by the Office of the Inspector General - Health and Human Services. The dataset contains individuals and entities excluded from federally funded healthcare programs due to reasons such as convictions for Medicare or Medicaid fraud. The file used in this case study represents the most recent complete dataset as of September 10, 2025, with monthly updates. The downloaded data covers the period from July 1, 1977, to September 18, 2025. The data are publicly available through the federal OIG website. Updates are made by Medicaid and Medicare fraud units based on criminal conviction cases. Reinstated individuals are excluded. The dataset is reliable for criminal convictions, though civil fraud cases are generally not listed. Not all providers have NPIs, and entities may have multiple exclusion dates. The granularity is at the individual or entity level.

CMS Open Payments – General Payments

The General Payments datasets are published by the Centers for Medicare and Medicaid Services and report financial relationships between drug and medical device manufacturers and covered recipients, including physicians and teaching hospitals. Payments include items such as meals, travel, research funding, gifts, and speaking fees. All entities are required to submit payments of \$10 or more yearly. The data used for this case study covers the period from January 1, 2021, through December 31, 2023, and was last updated on June 30, 2025 (annually updated). Data are publicly accessible via CSV download or API through the CMS Open Payments website. Payment records are validated and matched to providers before publication. NPIs are included in data from 2021 onward. Reporting entities face financial penalties for failure to report. The dataset is organized at the individual payment record level.

CMS Open Payments – Ownership and Investment Interests

This dataset, also published by CMS, contains information on ownership or investment interests held by physicians or their immediate family members in companies that manufacture drugs, medical devices, biologicals,

or medical supplies. The data covers the period from January 1, 2021, through December 31, 2023, with annual updates, the last update being on June 30, 2025. The data is publicly available via the CMS Open Payments portal. Information is reported annually by manufacturers and group purchasing organizations and undergoes validation and matching processes. Granularity is at the individual or entity ownership-interest level. Some records may have missing or incomplete NPIs, particularly in older data.

Medicare DMEPOS – By Supplier

This CMS dataset provides aggregated utilization, charges, Medicare allowed amounts, and Medicare payments for durable medical equipment, devices, and supplies, organized by supplier NPI. The data excludes claims associated with fewer than 10 beneficiaries to protect privacy. The dataset was last modified on June 26, 2025, and covers data from 2021 through 2023. The dataset was derived from administrative claims in the CMS Chronic Condition Data Warehouse (CCW) and includes provider details from the National Plan & Provider Enumeration System (NPPES). The dataset contains only final-action claims and is considered accurate and complete. The granularity is supplier-level aggregation.

Medicare DMEPOS – By Supplier and Service

This dataset further extends the supplier-level DMEPOS data by including Healthcare Common Procedure Coding System (HCPCS) codes and supplier rental indicators. It was last modified on September 9, 2025, and covers the period from 2021 through 2023. The dataset was derived from CMS CCW administrative claims and incorporates the Restructured BETOS Classification System for higher-level service grouping. Records involving fewer than 10 beneficiaries were not included. Compared to the supplier-only dataset, this file is more granular, reporting utilization and payments for each HCPCS code billed by a supplier.

Medicare DMEPOS – By Referring Provider

This dataset contains aggregated information on utilization, charges, Medicare allowed amounts, and Medicare payments organized by referring provider NPI and HCPCS code. It is derived from 100% final-action Part B non-institutional DMEPOS claims for Medicare fee-for-service beneficiaries in the CMS CCW for the 2021–2023 reporting periods. Provider demographic details are sourced from NPPES and extracted after the reporting year. Records derived from 10 or fewer beneficiaries are excluded. The granularity is provider-level aggregation.

Medicare DMEPOS – By Referring Provider and Service

This dataset further extends the referring provider data by incorporating the HCPCS code and supplier rental indicator. It was last modified on September 9, 2025, and covers the period from 2021 through 2023. Data are derived from CMS CCW final-action claims and include provider details from NPPES. Beneficiary-level

privacy protections are maintained through suppression rules. The granularity is provider-and service-level aggregation.

U.S. Census Bureau- County Tract Data

Demographic and socioeconomic context is provided by the U.S. Census Bureau's American Community Survey (ACS), in which each release represents a rolling 5-year estimate. The data used for this case study include total population, age and sex distribution, income, education attainment, poverty rates, race and ethnicity, and household and family composition at the county level. This data is publicly available through the Census Bureau website and is anonymized, aggregated, and methodologically documented, with disclosure avoidance procedures applied to ensure confidentiality in small geographic areas.

Medicare Enrollment

The Medicare Monthly Enrollment dataset reports up-to-date monthly counts of Medicare beneficiaries with hospital/medical coverage and prescription drug coverage across multiple geographies (national, state/territory, and county). Hospital/medical enrollment is further categorized by delivery model (Original Medicare vs. Medicare Advantage/Other Health Plans), and prescription drug enrollment can be analyzed by plan type (stand-alone Prescription Drug Plans vs. Medicare Advantage Prescription Drug plans). It supports both month-to-month and year-over-year enrollment trend analysis. For this project, the data were prefiltered to years 2021, 2022, and 2023, and we retained only total county-level enrollment by year.

New Data being incorporated: National Plan & Provider Enumeration System

The National Plan & Provider Enumeration System (NPES), maintained by the Centers for Medicare and Medicaid Services (CMS), is used in this case study as a provider reference dataset, providing standardized NPIs, provider types, activation statuses, and basic demographic information. This data was directly acquired from CMS's official government website and is freely available for research and administrative use.

Database:

Kindly refer to the following link for the [Entity-Relationship Diagram](#).

The ERD and database are still being developed and will go through a few more revisions during the winter break, with finalization expected before the start of the semester.

The relational database was developed using a star-schema-inspired framework consisting of fact and dimension tables that can support multi-year analysis without duplicating information. The core fact tables capture DMEPOS utilization and CMS Open Payments transactions, while dimension tables provide a reference point for providers, services (HCPCS), suppliers, dates, locations, census demographics, and OIG exclusion status.

The CMS-derived fact tables are connected through the National Provider Identifier (NPI) and year, allowing for the integration of provider-level utilization, referral, and financial data. HCPCS codes link services across DMEPOS datasets, while ZIP codes and county tracts sets up connections to census socioeconomic data. The OIG Exclusion List is linked using NPIs when available. Entity-level exclusions are recorded separately if NPIs are missing.

Storage and Management :

All cleaned, feature-engineered, and integrated datasets, along with the associated code, are stored in a dedicated directory on Hellbender, the University of Missouri's High-Performance Computing (HPC) cluster, as well as in the shared DSA JupyterHub case study folder.

Due to issues faced when running our code on JupyterHub, all future development and analysis will be conducted on Hellbender. The work will be stored within our advisor's directory, which is configured with restricted permissions. As a result, only authorized individuals have access to the folder, ensuring that all data and code are securely stored and appropriately controlled.

The directory location path in Hellbender: `//cluster/pixstor/haithcoatt-lab/f25cCaseStudy_Team2`

Data Shaping and Carpentry

Office of Inspector General (OIG) Exclusion List

The preparatory work for the LEIE (List of Excluded Individuals/Entities) dataset involves cleaning and standardizing the raw exclusion data downloaded from the OIG website, which may contain multiple rows per entity due to variations in name descriptions, addresses, exclusion dates, and types. The cleaning process begins by converting all column names to snake_case and applying appropriate data types to specified columns. Blank columns are dropped, and the data is filtered to retain only records with exclusion dates after 2020. To ensure a single row per NPI, only the most recent name description and most recent exclusion record per entity are retained. Additional filtering is applied to keep only specified exclusion types, as defined in a crosswalk stored in the team's Capstone folder.

The script also engineers new columns to support quality assessment and enrichment, including the number of exclusion dates, number and list of unique exclusion types, and the number of unique addresses per entity. A fraud_flag column is added and set to 1 for all rows. Two output files are generated: one that retains records with null NPIs for potential backfilling (leie_with_null_npi_clean.csv), and a final cleaned dataset with only valid NPIs and exclusions after 2020 (leie_with_valid_npi_clean.csv), both saved to /dsa/groups/casestudycf25/team02/.

CMS Open Payments – Ownership Payments

The preparatory work for the CMS Open Payments Ownership Payments dataset focuses on cleaning, standardizing, and aggregating physician ownership and investment interest information to support specialty-level fraud risk analysis. Raw ownership payment files covering program years 2021 through 2023 are ingested and standardized by converting all column names to snake_case and enforcing appropriate data types using the CMS Open Payments data dictionary. Records with missing or invalid physician National Provider Identifiers (NPIs) are removed to ensure reliable linkage across datasets.

Data quality assessment includes identifying and removing duplicate records based on physician NPI, applicable manufacturer or GPO identifier, and program year. Ownership-related monetary fields, including total_amount_invested_us_dollars and value_of_interest, are validated to ensure non-negative values and consistent units. Textual fields describing ownership terms and the nature of the interest are cleaned and normalized to reduce variability arising from free-text reporting.

Following cleaning, the dataset is transformed from a record-level structure to a specialty-level analytical table. Ownership indicators are aggregated by provider specialty, including counts of physicians with ownership interests, counts of distinct manufacturer or GPO ownership relationships, total and average ownership investment

values, and the proportion of ownership interests held directly by physicians versus immediate family members. “Dispute_status_for_publication” is retained as a binary indicator to capture contested ownership records that may reflect reporting irregularities or compliance risk.

The resulting ownership dataset provides interpretable, specialty-level indicators of financial entanglement and potential conflicts of interest. These features are designed to integrate directly with specialty-level DMEPOS utilization patterns, LEIE exclusion outcomes, and general payment signals, supporting downstream fraud risk diagnostics and comparative analysis across specialties.

CMS Open Payments – General Payments

The preparatory work for the CMS General Payments dataset involves integrating, cleaning, and transforming raw data files using a script intended to be run locally. The process begins by appending multiple years of data using the Polars library, specifically filtering for records where at least one of the following fields is marked “Yes”: `drug_or_biological` or `device_or_medical_supply`. This ensures that only records associated with relevant medical products are included. Once filtered, the data from all years is combined, duplicate records are removed, and a smaller raw output file is saved to the designated team directory for downstream use.

A comprehensive data cleaning and quality assessment process follows. All column names are standardized to `snake_case`, and data types are corrected using an accompanying data dictionary. Records with missing values in either the `total_amount_of_payment_us_dollars` or `covered_recipient_npi` fields are removed to maintain data completeness. ZIP codes are standardized by stripping digits after the dash, unrelated columns are dropped, and text fields are cleaned for consistency. Non-physician practitioners are excluded from the dataset, and missing product names are backfilled to improve completeness.

After cleaning, the dataset is transformed and normalized into domain-specific dimension tables to facilitate structured analysis. These include a fact table (`general_payments-sliced_clean.csv`), a product information table with one product ID (PDI) per row (`general_payments_products_clean.csv`), a provider table with one NPI per row (`general_payments_providers_clean.csv`), and a manufacturer-related table (`general_payments_manufacturers_clean.csv`). Each of these retains only the most recent program year for its respective domain. In addition to the dimension tables, a complete and cleaned version of the CMS General Payments dataset is returned in chunked CSVs, preserving the full set of records prior to dimensional splitting.

Medicare Enrollment

The preparatory work for the Medicare Enrollment dataset, using the most recent CMS download from June 2025 (covering years 2021 to 2023), focuses on cleaning, integrating, and standardizing the data for analytical use. The script begins by converting all column names to `snake_case` and applying data type corrections

using an accompanying data dictionary. The dataset is filtered to retain only total enrollment figures at the county and year level, with records from unknown counties removed. A key quality assessment step includes imputing missing beneficiary enrollment values: if a county has data for at least two years, the median value is used; otherwise, a static value of 5 is assigned to account for entries suppressed due to being fewer than 10. The final cleaned dataset is saved to /dsa/groups/casestudycf25/team02/medicare_enrollment_clean.csv for downstream use.

Census Data

The preparatory work for the U.S. Census dataset is designed to enrich specialty-level Medicaid and DMEPOS fraud analysis with county-level socioeconomic context for the period 2021 through 2023. Raw Census data are standardized by converting all column names to snake_case, applying consistent data types, and retaining only variables relevant to healthcare utilization, economic vulnerability, and access disparities.

Data cleaning includes handling missing and suppressed values using median imputation where appropriate to maintain coverage across counties. All geographic identifiers are normalized using county-level FIPS codes to support consistent joins with Medicare Enrollment data and specialty-level aggregates derived from provider ZIP codes. ZIP code-based information is mapped to counties using a crosswalk to ensure alignment across datasets.

A comprehensive set of Census-derived metrics is retained for analysis, including median household income, poverty rate, unemployment rate, uninsured rate, and population density. These variables are calculated and retained at the county level and are used as contextual indicators rather than direct fraud labels. The metrics are intended to capture structural and socioeconomic conditions that may influence healthcare access, utilization intensity, and billing patterns.

The cleaned Census dataset is integrated with Medicare Enrollment and specialty-level Open Payments aggregates to enable comparative analysis of fraud risk signals across counties with differing socioeconomic profiles. By incorporating Census context, the analysis supports more nuanced interpretation of specialty-level risk signals and reduces the likelihood of misattributing structurally driven utilization patterns to fraudulent behavior.

Medicare DMEPOS Provider

The CMS aggregates DMEPOS claims data by referring provider and by referring provider and service for each program year. Carpentry performed datasets with both levels of aggregation including manually appending labels for the program year and imputing null values. Fields in which missing values were deemed appropriate (such as middle initial or second line of street address) were left unmodified. Null credential records

were replaced by an empty string, and text normalization to lowercase and remove punctuation was performed. Missing Rural-Urban Commuting Area (RUCA) data was imputed based on zip code mapping from the 2010 RUCA codes published by the U.S. Department of Agriculture. Missing values for total supplier beneficiaries were imputed as five following the methodology of Khoshgoftaar & Johnson (2018).

DMEPOS – By Referring Provider & Service

The preparatory work for the DMEPOS – By Referral Provider & Service involves integrating, cleaning, and transforming data files using a script intended to be run locally for analytics purposes. The script uses “mup_dme_ry25_p05_v10_dy21_rfrhpr.csv“, “mup_dme_ry25_p05_v10_dy22_rfrhpr.csv“, and “mup_dme_ry25_p05_v10_dy23_rfrhpr.csv“, which are stored in the path /dsa/groups/casestudycf25/team02/. The original files included 1,516,153, 1,457,378, 1,439,587 for a total of 4,413,118 records in the combined dataset.

Two independent runs of data carpentry were performed in this dataset. In one, the preparatory work involved loading and combining the 2021–2023 “DMEPOS - By Referral Provider” datasets, assessing data quality, and identifying suppressed values. String columns were standardized by converting to lowercase, removing special characters, and replacing spaces with underscores. Additional indicators were added: “DME_Sprsn_Ind” to mark suppressed entries, “Excluded” to flag excluded cases, and “Supplier” to indicate if the provider also acted as a supplier. This cleaned dataset was saved to /dsa/groups/casestudycf25/team02/DMEPOS_RefPro_Ser_nar_clean.csv.

From this dataset, other datasets , such as “DMEPOS_cleaned_grouping.csv”, “DMEPOS_cleaned_serv.csv”, and “DMEPOS_Amount_Stats_labeled.csv” were derived.

- “DMEPOS_cleaned_grouping.csv” contains every unique combination of RBCS levels, ID, and description, HCPCS code, and description.
- “DMEPOS_cleaned_serv.csv” contains every unique combination of HCPCS code and description.
- “DMEPOS_Amount_Stats.csv” contains NPI-level aggregated statistics (mean, sum, median, standard deviation, min, and max) for all amount-related metrics, along with an exclusion flag highlighting NPIs that are excluded.

These datasets are stored within the shared team space: /dsa/groups/casestudycf25/team02/.

In a separate data carpentry run, the procedures described prior to this section’s head were performed in addition to carpentry on the referring provider specialty codes (Rfrg_Privr_Spclty_Cd) field which was unique to the referring provider and service datasets. Missing codes were imputed based on the best match to referring provider specialty description from the Medicare Provider and Supplier Taxonomy Crosswalk published by CMS

as deemed by personal judgement. If the descriptions were deemed too vague or unclear, the specialty code was imputed as an empty string. The final, clean dataset from this run was written out to the path /dsa/group/casestudycf25/team02/ DMEPOS_rfrhpr_clean.csv.

From this file the following datasets were subsequently derived:

- DMEPOS_rfrhpr_clean_labeled.csv
- DMEPOS_Amount_Stats_labeled.csv

For the combined, cleaned dataset, we determined that the data fields Rfrg_NPI, HCPCS_CD, Suplr_Rentl_Ind, and Year composed a candidate key. We also interrogated further dependencies to determine how the dataset can be reduced for efficient storage. We found a substantial quantity of HCPCS codes associated with two RBCS codes and confirmed the relationship was not an artifact of whether the service equipment was rented. However, we confirmed all data fields prefixed by "RBCS" were determined entirely RBCS_Id field. We determined that fields prefixed by "Rfrg" are dependent on both Rfrg_NPI and Year; that is, referring provider attributes associated with a particular NPI number are subject to change from year to year. Additionally, we observed that none of cities, states, FIPS codes, nor RUCA are entirely determined by zip code; that is, there are zip codes crossing city and state boundaries and occupying multiple RUCA designations.

This dataset is the basis for the feature-engineered dataset (DMEPOS_Amount_Stats_labeled.csv) within which a new set of features were calculated as the mean, median, minimum, maximum, and standard deviation of numeric data fields for each unique NPI aggregated by program year. We also engineered an additional two features, Tot_Suplr_Nonrentl_HCPCS_Cds and Tot_Suplr_Rentl_HCPCS_Cds, corresponding to the count of unique HCPCS codes with the negative and positive rental tag respectively for each unique NPI aggregated by the program year. This process reduced the quantity of records in the dataset to 879,531.

DMEPOS - By referring provider

The preparatory work for the DMEPOS – By Referral Provider & Service involves integrating, cleaning, and transforming data files using a script intended to be run locally for analytics purposes. The script uses “mup_dme_ry25_p05_v10_dy21_rfr.csv”, “mup_dme_ry25_p05_v10_dy22_rfr.csv”, and “mup_dme_ry25_p05_v10_dy23_rfr.csv”, which are stored in the path /dsa/groups/casestudycf25/team02/. The original files included 384,264, 385,637, and 387,352 unique records respectively for a total of 1,157,253 records in the combined dataset.

The fields which were unique to this dataset included claims data for durable-medical-equipment-specific (DME) claims, prosthetic-and-orthotic-suppression-specific (POS) claims, and drug-specific claims as well as beneficiary demographic information. We determined that 425,378 records had the total supplier beneficiaries entry suppressed and found that median values for non-null POS-specific and drug-specific claims data for such records were all zero, and as such, null values for these fields were imputed as zero. In the case of DME-specific

claims, such median values were nonzero and were used to impute missing data except for suppressed numbers of DME beneficiaries which were imputed as five to remain consistent with the methodology of imputing total supplier beneficiary counts. Of the 425,378 records for which the total number of beneficiaries was suppressed, all beneficiary demographic data was null for except for beneficiary average age and beneficiary risk-score. As such, missing demographic values were imputed as zero. After these procedures and including data cleaning describe previously, the final, clean dataset was written out to the path /dsa/group/casestudycf25/team02/DMEPOS_rfr_clean.csv.

From this file the following datasets were subsequently derived:

- DMEPOS_rfr_clean_labeled.csv
- DMEPOS_Amount_Stats_labeled.csv

We interrogated the relational dependencies among data fields in the combined, cleaned referring-provider-level claims data to assess possible reductions for efficient storage. We determined that the data fields Rfrg_NPI and Year comprise a candidate key for this dataset.

CMS DMEPOS Supplier

DMEPOS – By Supplier

The preparatory work for the DMEPOS – By Supplier dataset involves integrating, cleaning, and transforming data files using a script that is intended to be run locally for analytics purposes. The script uses “DMEPOS_suplr_2021.csv”, “DMEPOS_suplr_2022.csv” and “DMEPOS_suplr_2023.csv”, which are stored in the path /dsa/groups/casestudycf25/team02/. The cleaned file includes 198,621 unique entries and 43 attributes.

The process begins with a simple exploration. The structure and quality of the data were assessed, missing or suppressed values were identified, and string and numeric attributes were examined to ensure consistency. This was followed by the cleaning phase, where non-essential columns were removed from the data, while core metrics, such as supplier demographics, total claims, beneficiaries, submitted charges, and risk scores, were retained. Additionally, a Year column was added, the column order was rearranged, and data from the three years were combined into a single dataset. String columns were cleaned, converted to lowercase, and formatted for consistency. Suppression flags were mapped to ‘y’ or ‘n’, to indicate suppression of data and numeric suppressed values were replaced with a static value of 5 as a placeholder to preserve records.

The script then saves the dataset under the name “DMEPOS_cleaned_suplr.csv” within the shared team space; /dsa/groups/casestudycf25/team02/.

DMEPOS – By Supplier & Service

The preparatory work for the DMEPOS – By Supplier & Service dataset involves integrating, cleaning, and transforming data files using a script that is intended to be run locally for analytics purposes. The script uses “DMEPOS_Supplier_Service_2021.csv”, “DMEPOS_Supplier_Service_2022.csv”, and “DMEPOS_Supplier_Service_2023.csv”, which are stored in the path /dsa/groups/casestudycf25/team02/. The cleaned file includes 1,218,152 entries and 32 attributes.

Similar to the previous carpentry, the process begins with a simple exploration, where the data structure, quality, and data types of string and numeric attributes were assessed, followed by the identification of missing or suppressed values. Unlike the DMEPOS – By Supplier dataset, this data did not have many gaps. The suppressed values only affected the "Tot_Suplr_Benes" column. For the cleaning phase, a year column was added, and all three datasets were combined into a single dataset. RUCA-related columns ('Suplr_Prldr_RUCA_Cat', 'Suplr_Prldr_RUCA', 'Suplr_Prldr_RUCA_Desc') were removed. String columns were standardized by converting them to lowercase, removing special characters, replacing spaces with underscores, and filling empty or null values with ‘na’. Additionally, a suppression indicator, “DME_Sprsn_Ind”, was added to mark entries that were suppressed.

The script then saves the dataset under the name “DMEPOS_cleaned_suplr_serv.csv” within the shared team space: /dsa/groups/casestudycf25/team02/.

Provider Specialties

The preparatory work for the specialty information dataset involves consolidating provider specialty data into a single, standardized file with one row per NPI reflecting the most recent specialty. This process integrates data from two sources: the DMEPOS Referring Provider dataset and the General Payments provider data. The script uses /dsa/groups/casestudycf25/team02/general_payments_providers_clean.csv and /dsa/groups/casestudycf25/team02/DMEPOS_rfr_clean.csv as inputs. Data cleaning steps include selecting only one row per NPI from the DMEPOS dataset, prioritizing the most recent and NPPES-sourced records, and retaining only the most recent specialty information from the General Payments dataset. These interim outputs are saved as dmepos_specialties_unique.csv and gp_specialties_unique.csv. A manually created crosswalk file, specialty_crosswalk_gpt_backfill.csv, is then used to deduplicate and backfill specialty hierarchy information. This crosswalk, initially generated with ChatGPT assistance and manually validated, maps original specialty names to standardized groupings. The final merged output, which includes one specialty per NPI with hierarchy details, is saved as /dsa/groups/casestudycf25/team02/one_specialty_per_npi.csv.

Unified Dataset

The preparatory work for the `unified_dataset.csv` focuses on creating a comprehensive, integrated dataset specifically at the NPI and program year level. This script merges multiple cleaned datasets—including `general_payments_products_clean.csv`, `general_payments-sliced_clean.csv`, `ownership_payment_clean.csv`, `DMEPOS_Amount_Stats_labeled.csv`, and `one_specialty_per_npi.csv`—to align all data to this consistent grain. The process begins by combining the two Open Payments datasets, followed by sequential merges with the DMEPOS Referring Provider data, the LEIE exclusion list, and specialty hierarchy information. To ensure data integrity, all joins are strictly controlled using `validate="one_to_one"` to avoid cross joins or duplicate records.

Key data cleaning and transformation steps include converting column names to `snake_case`, mapping Yes/No values to binary indicators, and pivoting the General Payments data on `product_indicator` to generate device-type counts. Aggregations are performed to calculate payment counts, sums, and min/max dates, grouped by NPI, year, and manufacturer ID. Ownership data is similarly grouped and pivoted, then left-joined with the General Payments data, which is ultimately rolled up to the NPI-year level. The unified dataset is then left-joined with the DMEPOS data, ensuring all DMEPOS records are retained, followed by a conditional merge with the LEIE data and an inner join with the specialty dataset. The script outputs interim CSVs throughout the process, and the final result is the `unified_dataset.csv`, a structured dataset at the year/NPI level, saved to `/dsa/groups/casestudycf25/team02/`.

Exploratory Data Analysis and Visualizations

DMEPOS – By Referring Provider and Service

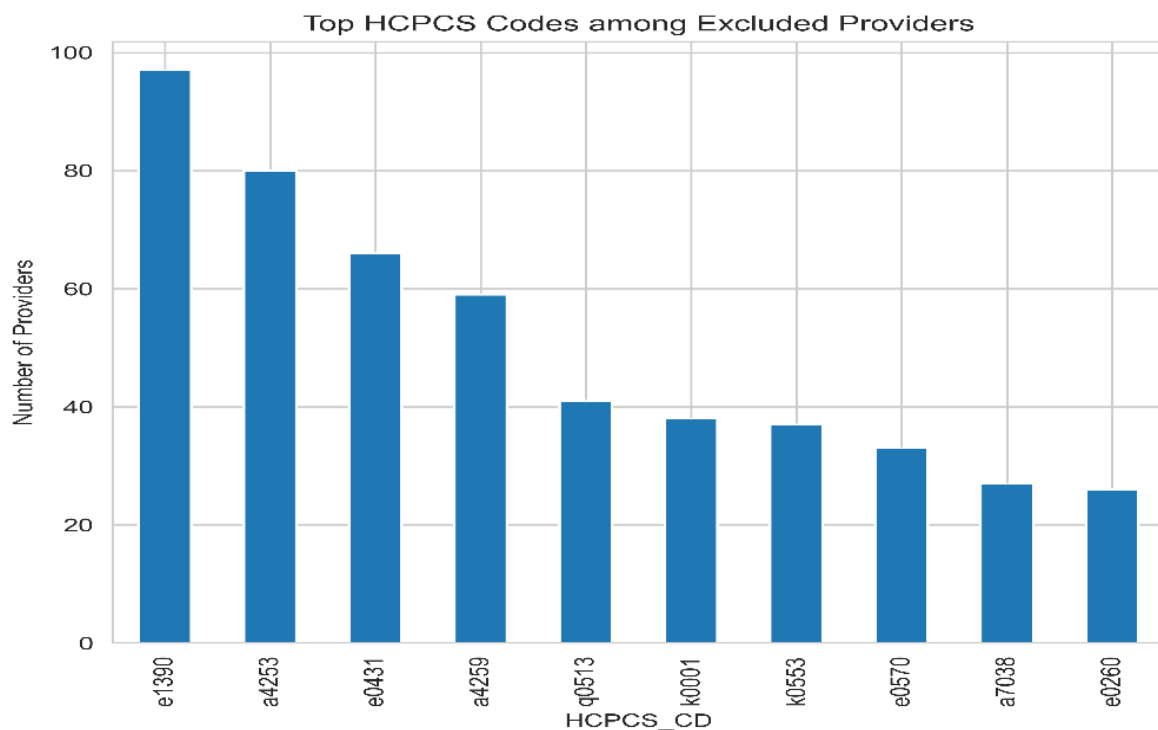
The EDA was conducted on a cleaned DMEPOS – By Referring Provider dataset, which includes NPI, HCPCS procedure codes, claim counts and amounts, geographic indicators, and a temporal variable. Additionally, an exclusion status attribute was created using the cleaned LEIE dataset to flag providers who are excluded.

Basic distribution of excluded and non-excluded claims:

- Total claims: 4,413,118
- Non-excluded provider claims: 4,411,705
- Excluded provider claims: 1,413

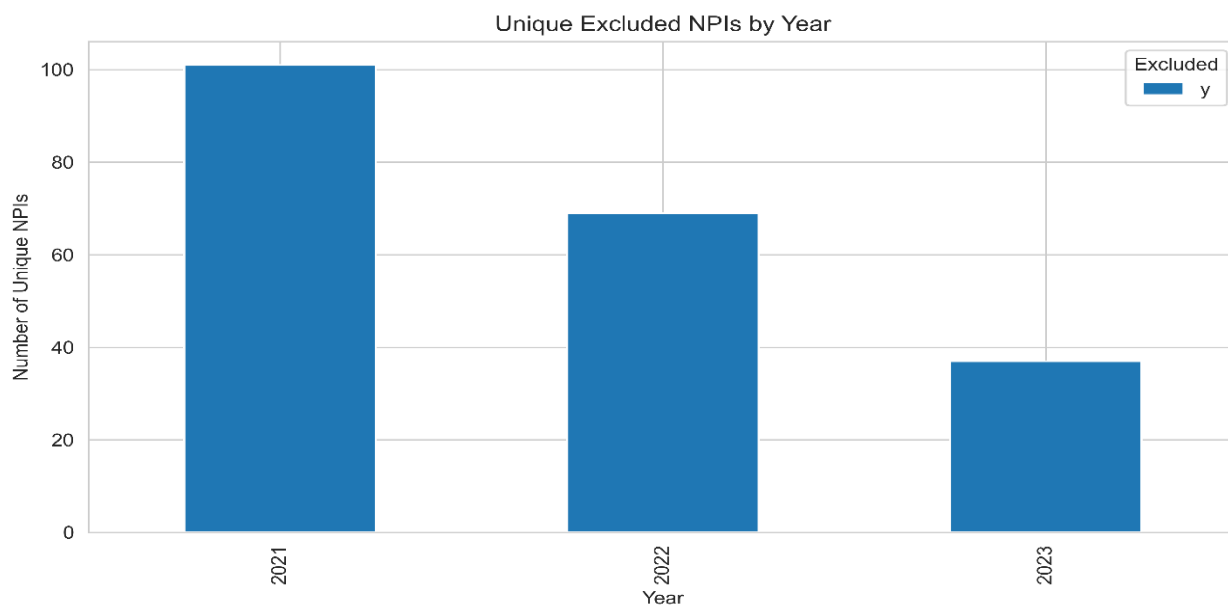
Due to the high class imbalance, the primary focus shifted to identifying patterns among excluded providers.

- Providers holding an MD (Doctor of Medicine) accounted for approximately 80% of the excluded cases.

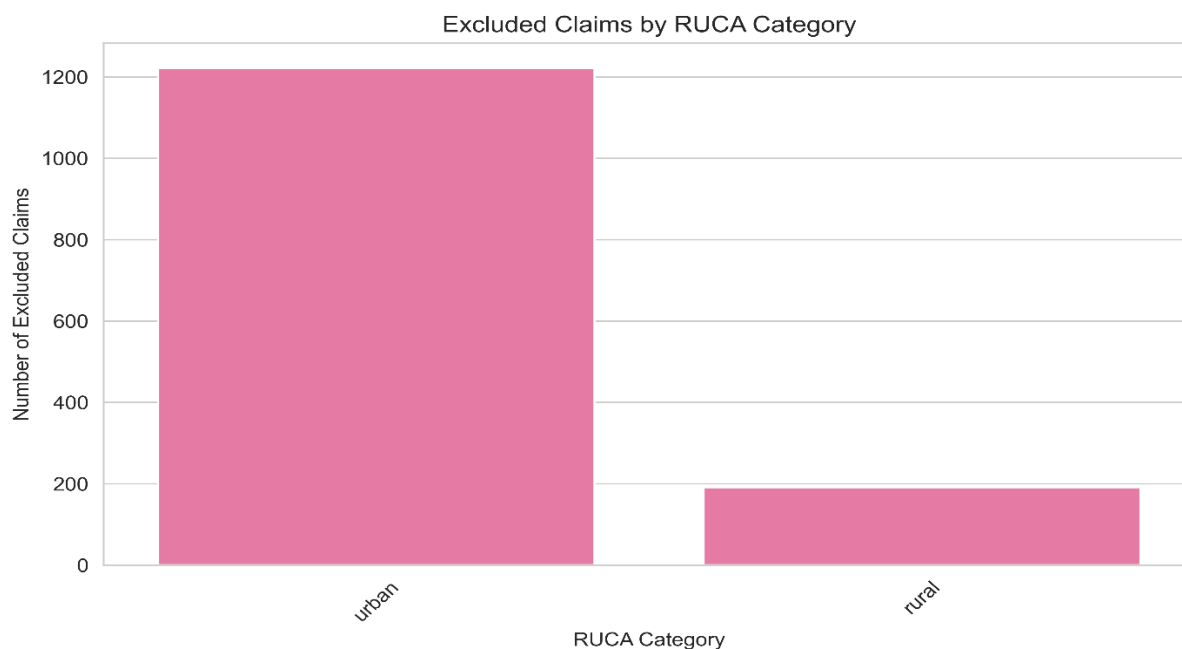


This visualization highlights the HCPCS codes most frequently associated with excluded referring providers. Most of these services are related to oxygen supply equipment, which are often billed monthly as rentals. This creates opportunities for fraud, either through upcoding or by failing to deliver the billed equipment.

Additionally, this equipment is often prescribed to elderly or chronically ill Medicare beneficiaries, who may not fully understand their bills or may be unaware of unnecessary or ongoing services.

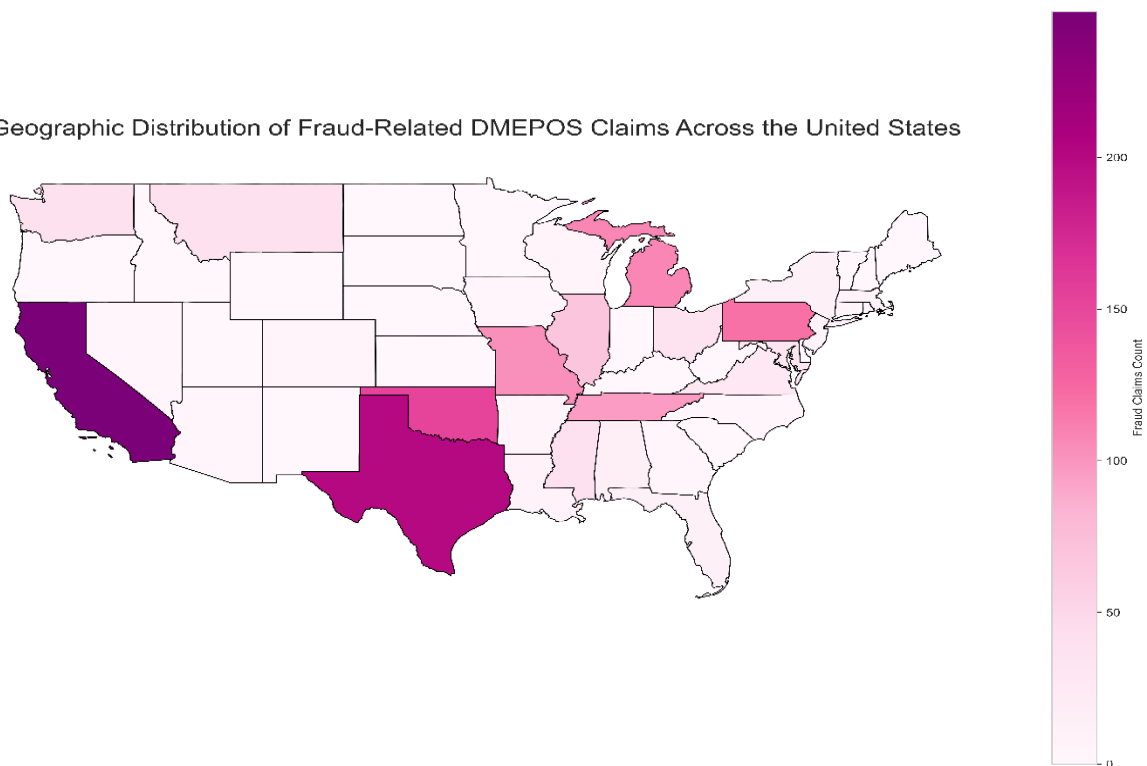


This visualization shows the number of unique referring providers excluded from Medicare participation over the 2021–2023 period. There is a steady decline in excluded NPIs during this time. This does not necessarily mean there is a decline in fraud, but rather that fewer unique providers are being excluded, which could result from changes in enforcement or provider compliance behavior. Regardless, this visualization establishes time as a critical contextual component for fraud analysis.



This visualization compares fraud-related excluded claims across urban and rural settings. Urban areas typically have higher provider density and claim volumes, which explains the higher fraud counts. While the type of area appears to influence these outcomes, it is not necessarily the cause.

Geographic Distribution of Fraud-Related DMEPOS Claims Across the United States



This visualization builds on the previous visual by displaying the state-level distribution of fraud-related claims, aiming to reveal potential clusters. California and Texas have the highest number of fraudulent claims, likely due to their large populations. Interestingly, they are followed by Oklahoma, Pennsylvania, and Michigan. When discussing with stakeholders, states like Ohio are often cited as fraud hotspots, so it is interesting to see a different story being told from this visual.

We calculated summary statistics on the cleaned, combined DMEPOS by Referring Provider and Service data as displayed in the following table.

	count	mean	std	min	25%	50%	75%	max
Tot_Suplrs	4,413,118	4.35	5.18	1.00	1.00	3.00	5.00	208.00
Tot_Suplr_Benes	4,413,118	14.57	38.78	5.00	5.00	5.00	12.00	8,172.00
Tot_Suplr_Clms	4,413,118	42.45	94.67	11.00	13.00	19.00	35.00	13,871.00
Tot_Suplr_Srvcs	4,413,118	1,263.49	18,244.36	11.00	16.00	34.00	139.00	5,402,250.00
Avg_Suplr_Sbmtcd_Chrg	4,413,118	209.19	554.40	0.00	19.25	69.69	190.00	77,612.10
Avg_Suplr_Mdcr_Alowd_Amt	4,413,118	75.43	263.02	0.00	8.02	23.05	62.38	27,701.77
Avg_Suplr_Mdcr_Pymt_Amt	4,413,118	57.67	206.43	0.00	5.31	17.11	46.31	21,718.19
Avg_Suplr_Mdcr_Stdzd_Amt	4,413,118	59.00	206.77	0.00	5.52	17.49	50.29	21,496.42

We additionally calculated summary statistics for the DMEPOS by Referring Provider and Service dataset within each class label.

Non-fraud summary statistics

cou nt	mea n	st d	mi n	25 %	50 %	75 %	max		
Tot_Suplrs	4,411,705.00	4.35	5.18	1.00	1.00	3.00	5.00	208.00	
Tot_Suplr_Benes	4,411,705.00	14.57	38.65	5.00	5.00	5.00	12.00	8,172.00	
Tot_Suplr_Clms	4,411,705.00	42.45	94.60	11.00	13.00	19.00	35.00	13,871.00	
Tot_Suplr_Srvcs	4,411,705.00	1,263.68	18,247.21	11.00	16.00	34.00	139.00	5,402,250.00	
Avg_Suplr_Sbmtd_Chrg	4,411,705.00	209.18	554.39	0.00	19.25	69.69	190.00	77,612.10	
Avg_Suplr_Mdcr_Alowd_Amt	4,411,705.00	75.42	263.01	0.00	8.02	23.05	62.38	27,701.77	
Avg_Suplr_Mdcr_Pymt_Amt	4,411,705.00	57.66	206.43	0.00	5.31	17.11	46.31	21,718.19	
Avg_Suplr_Mdcr_Stdzd_Amt	4,411,705.00	58.99	206.77	0.00	5.53	17.49	50.29	21,496.42	

Fraud summary statistics

	count	mean	std	min	25%	50%	75%	max
Tot_Suplrs	1,413.00	4.49	6.23	1.00	1.00	3.00	5.00	73.00
Tot_Suplr_Benes	1,413.00	35.30	178.68	5.00	5.00	5.00	13.00	2,987.00
Tot_Suplr_Clms	1,413.00	61.22	217.11	11.00	13.00	20.00	37.00	3,881.00
Tot_Suplr_Srvcs	1,413.00	683.93	2,681.03	11.00	17.00	39.00	219.00	63,000.00
Avg_Suplr_Sbmtd_Chrg	1,413.00	240.57	573.35	0.13	12.74	57.95	187.00	5,457.34
Avg_Suplr_Mdcr_Alowd_Amt	1,413.00	104.84	269.25	0.03	5.23	20.34	66.64	3,026.97
Avg_Suplr_Mdcr_Pymt_Amt	1,413.00	80.41	210.06	0.02	3.99	14.91	47.42	2,359.51
Avg_Suplr_Mdcr_Stdzd_Amt	1,413.00	81.48	214.98	0.02	4.24	15.56	50.29	2,696.02

After normalizing each feature by its standard deviation in the population, the differences in means between the fraud and non-fraud classes are most pronounced for the Total Supplier Services and Total Supplier Beneficiaries features where average is greater for the non-fraud class. However, this may be explained by the presence of outliers in the non-fraud class which the fraud class lacks as determined by further analysis.

	Nrmd_Sqrd_Class_Mean_Diff
Tot_Suplr_Srvcs	18.42
Tot_Suplr_Benes	11.09
Tot_Suplr_Clms	3.72
Avg_Suplr_Mdcr_Alowd_Amt	3.29
Avg_Suplr_Mdcr_Pymt_Amt	2.51

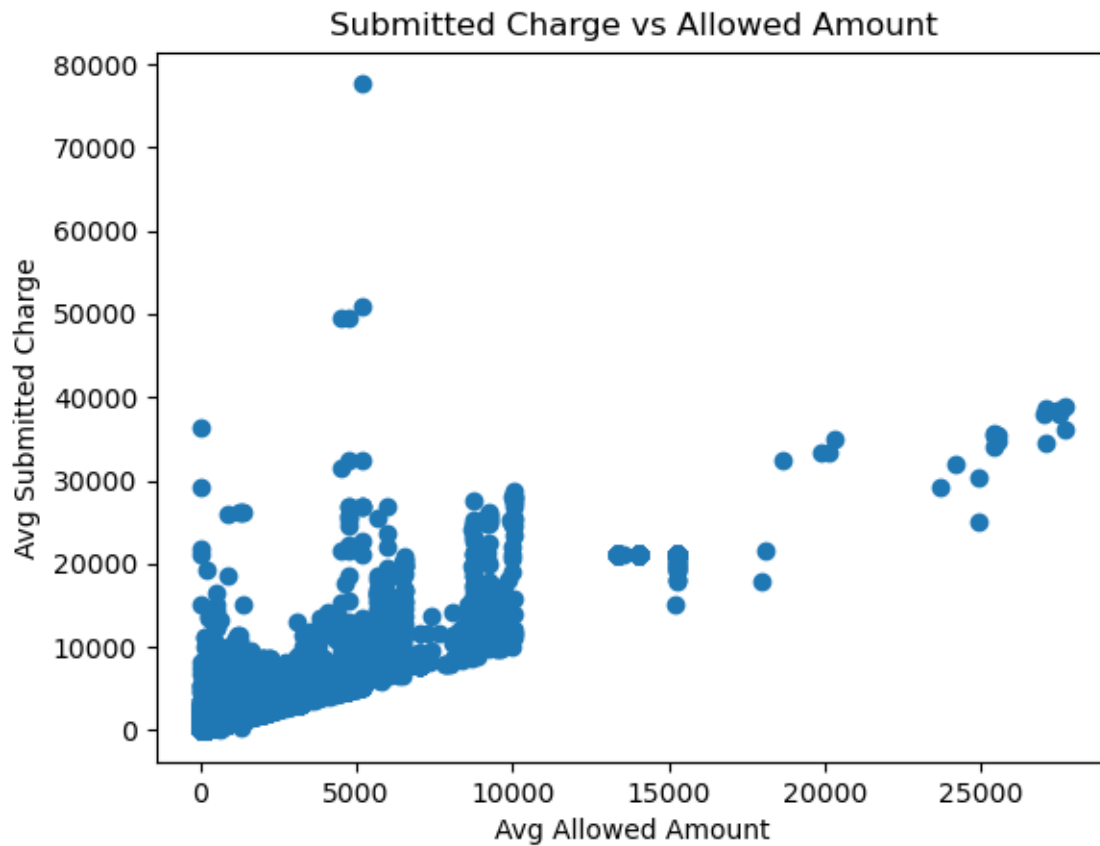
Avg_Suplr_Mdcr_Stdzd_Amt	2.45
Avg_Suplr_Sbmted_Chrg	1.78
Tot_Suplrs	0.00

We compute the absolute difference of medians of the classes divided by the population's inter-quartile range (Abs_Nrmd_Class_Median_Diff) which is not sensitive to outliers. By the normalized absolute difference in medians between classes, the typical amounts of Total Suppliers and Total Supplier Beneficiaries for fraudulent and legitimate providers are virtually indistinguishable. Instead, we find slightly lower Average Supplier Submitted Charges, Average Supplier Medicare Payment Amounts, Average Supplier Medicare Allowed Amounts, and Average Supplier Medicare Standardized Amounts for fraudulent providers as well as slightly higher Total Supplier Claims. We speculate that the submission of lower-value claims with a greater frequency is indicative of fraud.

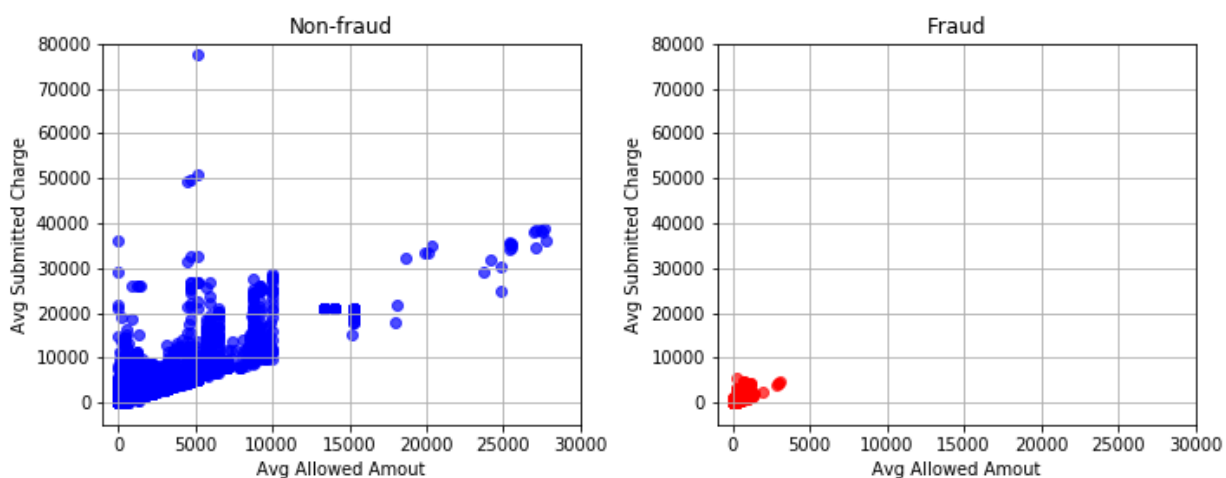
	Abs_Nrmd_Class_Median_Diff
Avg_Suplr_Sbmted_Chrg	0.07
Avg_Suplr_Mdcr_Pymt_Amt	0.05
Avg_Suplr_Mdcr_Alowd_Amt	0.05
Tot_Suplr_Clms	0.05
Avg_Suplr_Mdcr_Stdzd_Amt	0.04
Tot_Suplr_Srvcs	0.04
Tot_Suplrs	0.00
Tot_Suplr_Benes	0.00

We visually analyzed bivariate relationships comparing Average Supplier Submitted Charges (Avg_Suplr_Sbmted_Chrg) to Average Supplier Medicare Allowed Amount (Avg_Suplr_Mdcr_Alowd_Amt). We found a clear positive association between the two variables with some outliers both in terms of submitted charge and allowed amount. The outliers in submitted charge deviate from the general trend considerably, but the outliers in average submitted about appear to be commensurate with the approximately linear relationship between the

variables.

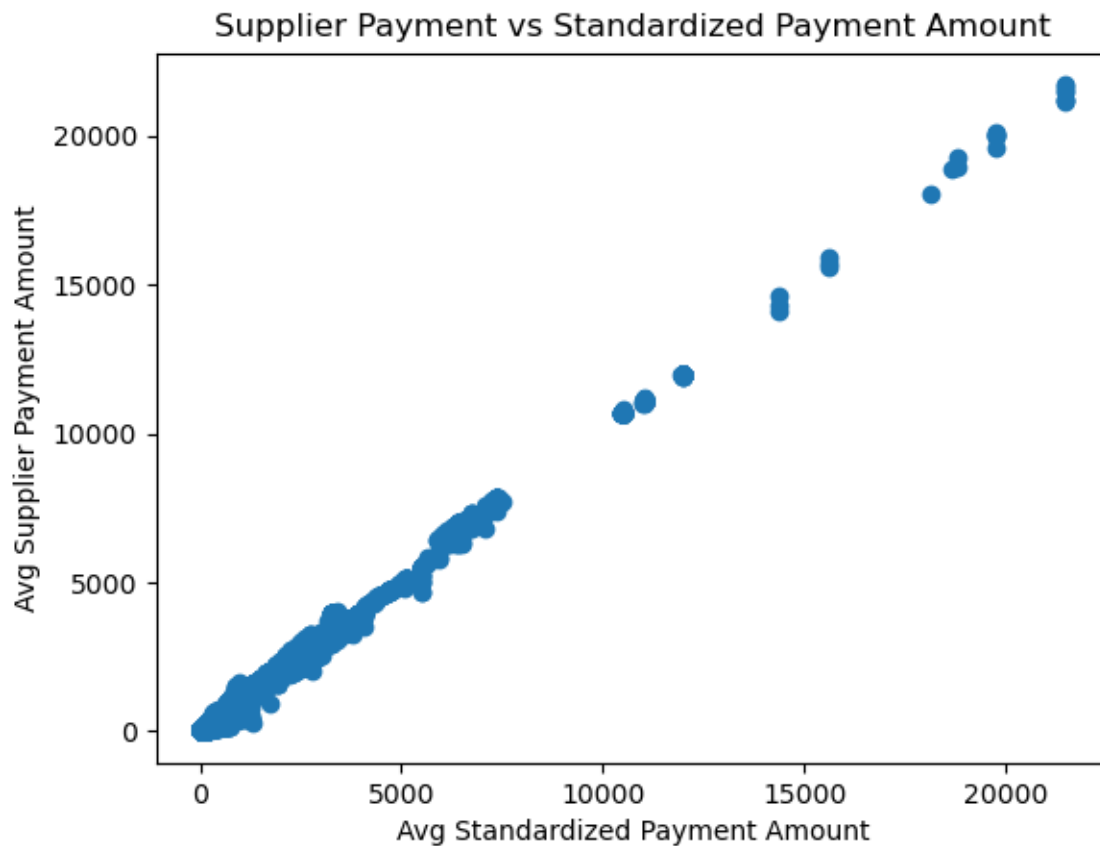


We segregated the data by fraud class and noted by visual inspection that outliers appear only in claims from non-fraudulent providers.

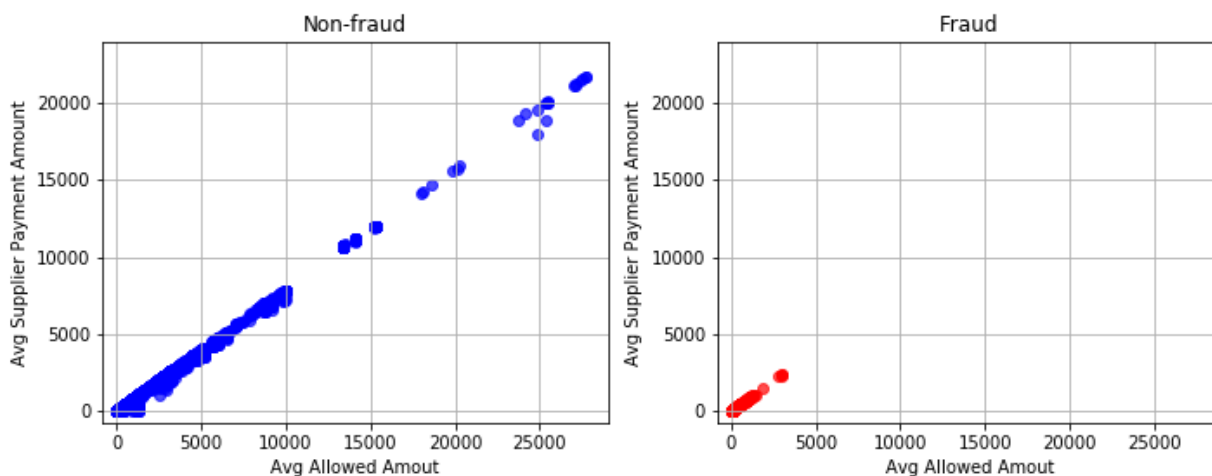


We also plotted Average Supplier Medicare Payment Amount (Avg_Suplr_Mdcr_Pymt_Amt) against Average Supplier Medicare Standard Payment Amount (Avg_Suplr_Mdcr_Stdzd_Amt) observing a nearly-perfect

linear relationship. This finding was unsurprising as we expected that amounts disbursed by Medicare would not deviate significantly from the standard amounts.

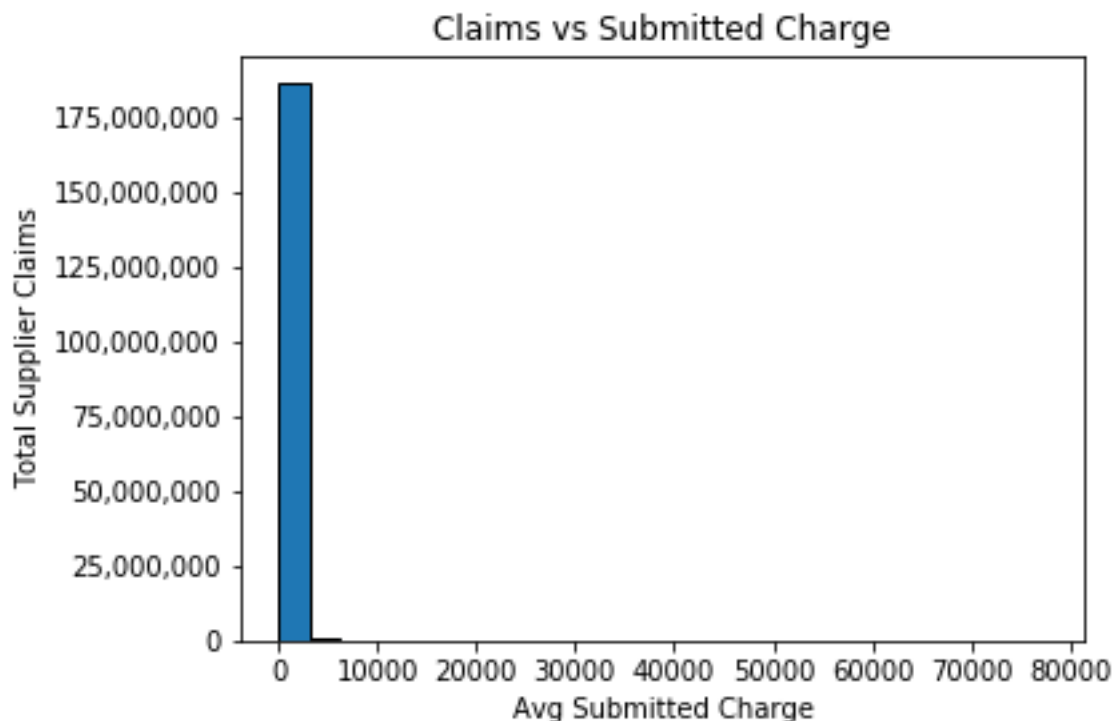


No differences between the fraudulent and non-fraudulent classes were visually apparent after segregating their data as evidenced by the following figure.

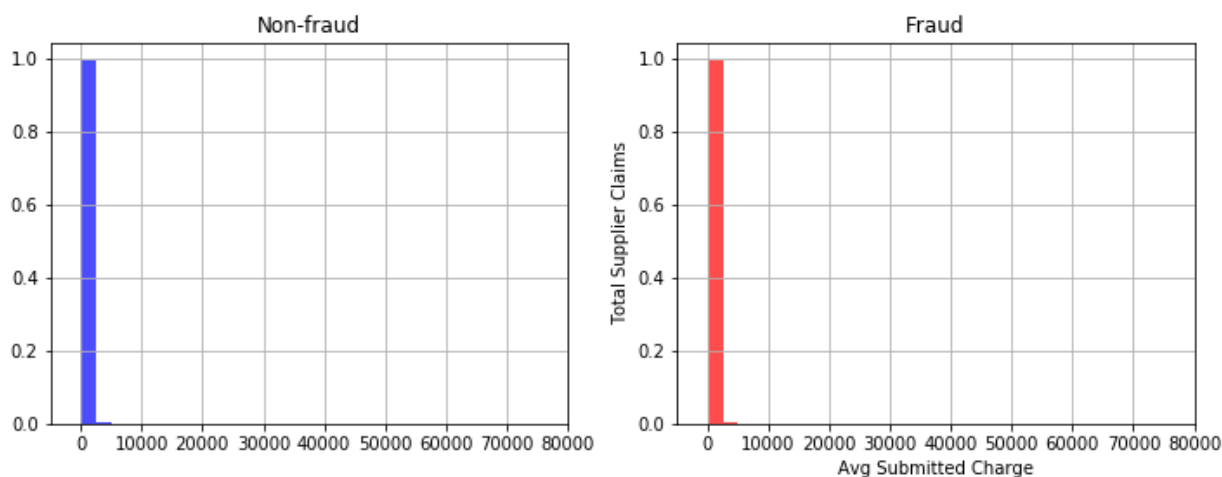


We binned Avg_Suplr_Sbmted_Chrg in increments of \$2,500 from \$0 to \$80,000 and constructed a histogram of total supplier claims submitted over all program years across all providers. We found that the total

number of claims for which the average supplier submitted charge was less than \$2,500 for the given service provided by a given referring provider was over 175 million.

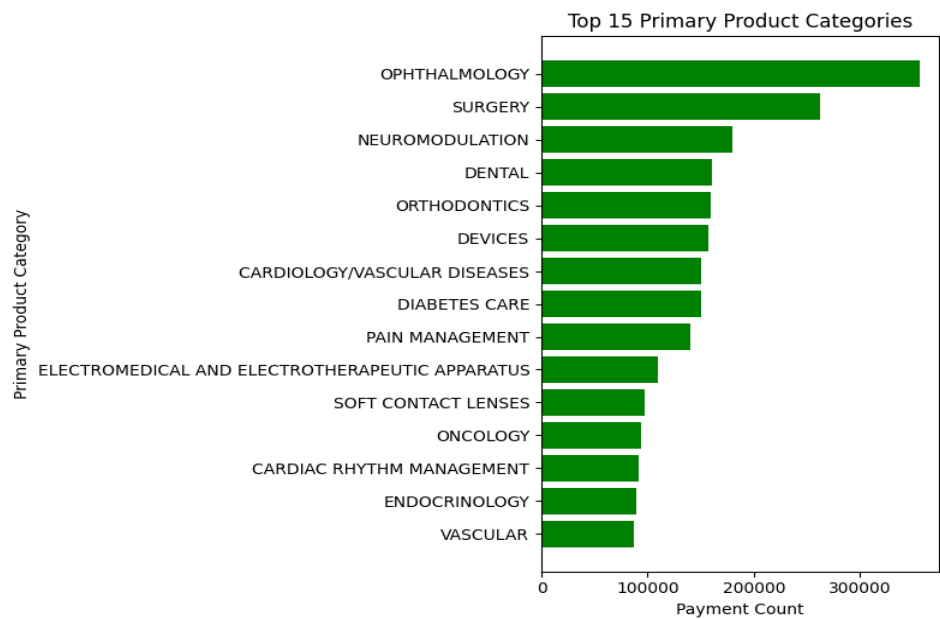


Given the severe class imbalance, when we segregated by fraud class, we elected to plot the histograms as relative frequencies of bins within each class as the total claims from non-fraudulent providers dwarfs that of fraudulent providers. We find that in both classes, nearly all submitted claims had an average supplier submitted charge by referring provider and service less than \$2,500.



General Payments

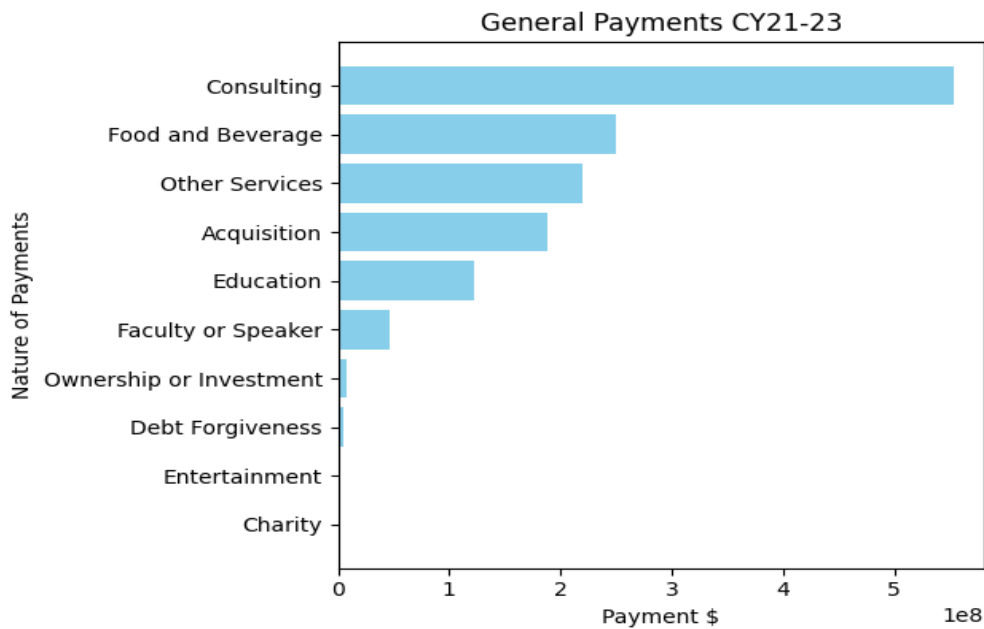
The EDA was conducted on a cleaned CMS General Payments dataset, which includes recipient NPIs, manufacturing IDs, product names, payment dates, and geographic indicators. Additionally, an exclusion status attribute was created using the cleaned LEIE dataset to flag providers who are excluded.



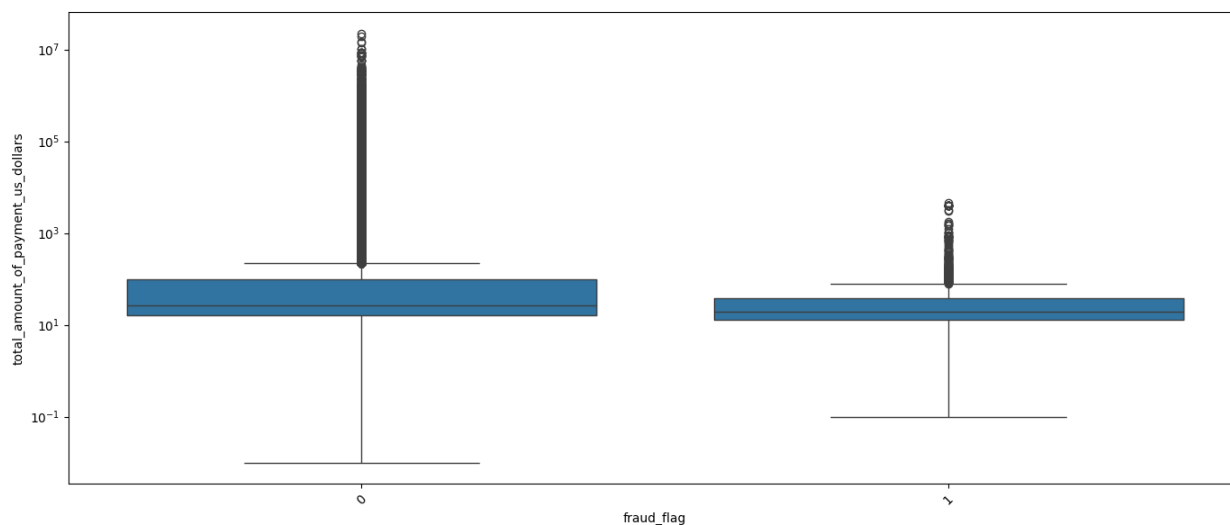
The bar charts in green and blue are based on the count and sum of payments from manufacturers and grouped by provider specialties and nature of manufacturers payments to recipient providers.

As the descriptive charts show, Opthamalogy is a top primary product category across CY2021-2023. Additionally, most payments are grouped in Consulting and Food &

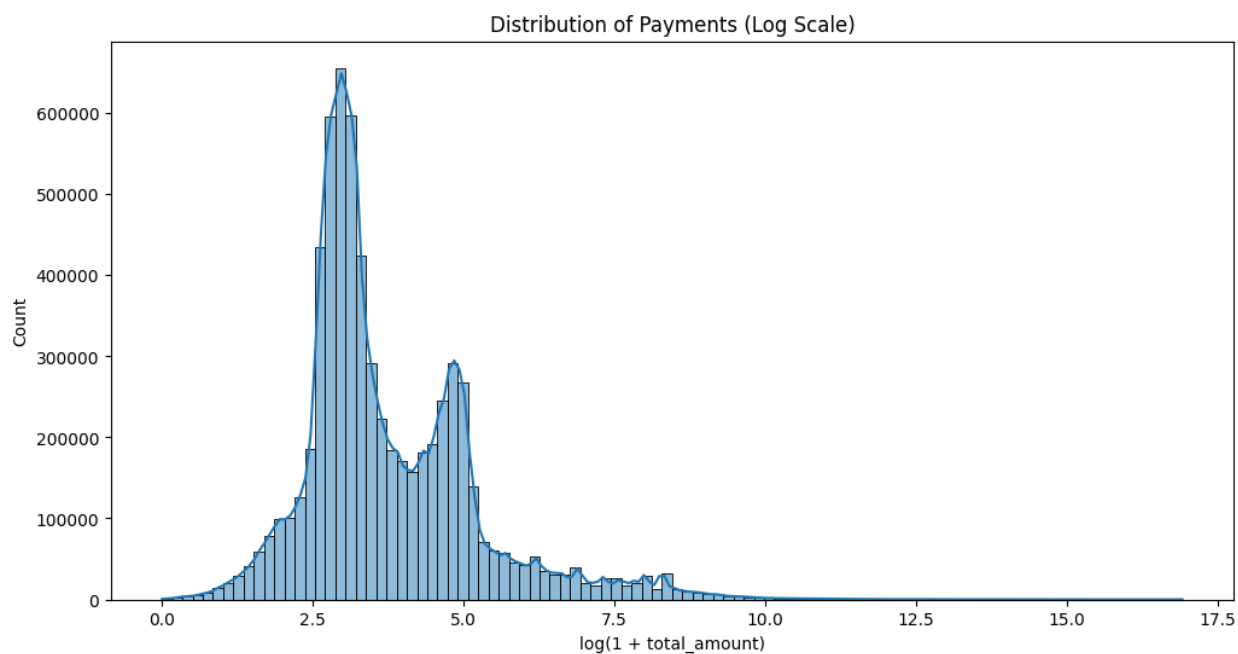
Beverage as the nature of a device manufacturer’s payments towards providers.



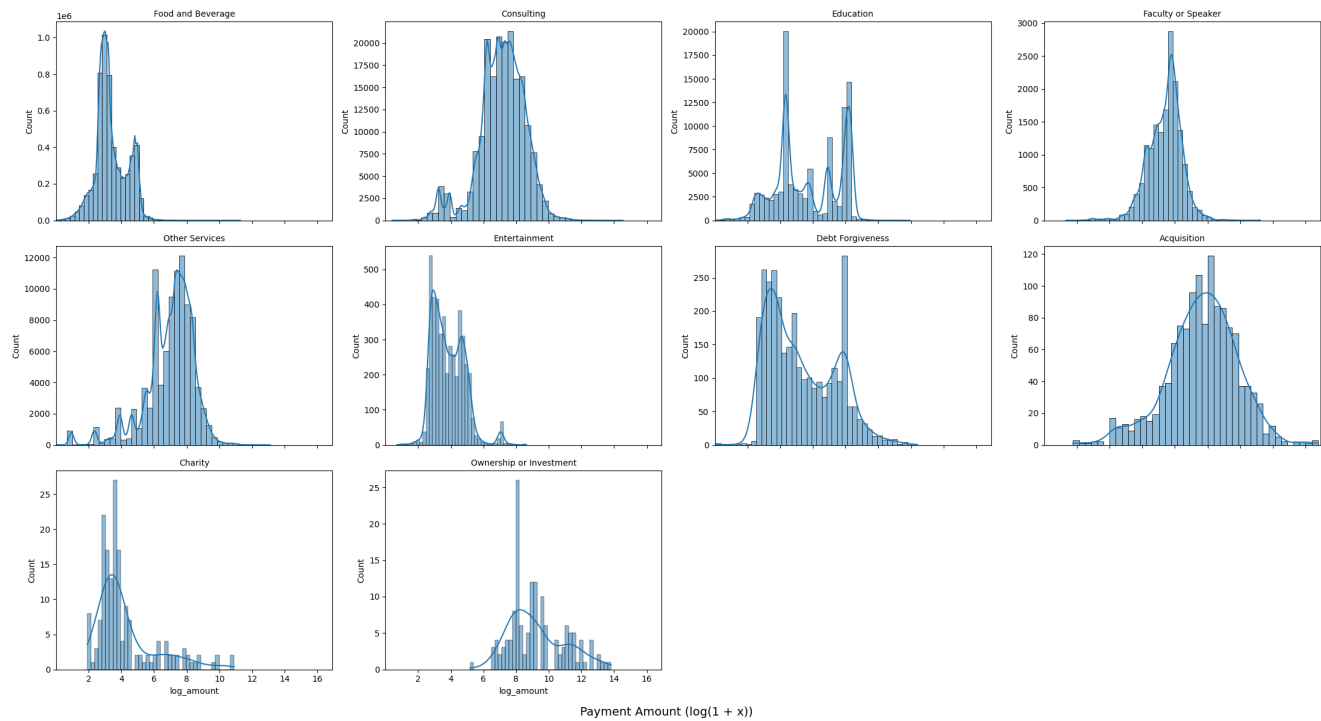
The blue adjusted boxplots below are based on the total payments made by manufacturers and separated by fraud class. There is no significant difference between median payments or distribution of payments between classes.



The histogram in blue is based on the total payments made by manufacturers across all three years. Interestingly, it is bimodal.



Bimodal, normal distributions, and a few skewed distributions from all payment transactions. In the provider risk scoring radar visual, the transactions that are outside of the normal distributions via z-score are considered higher risk. Further stats need to be run to correctly classify each distribution.



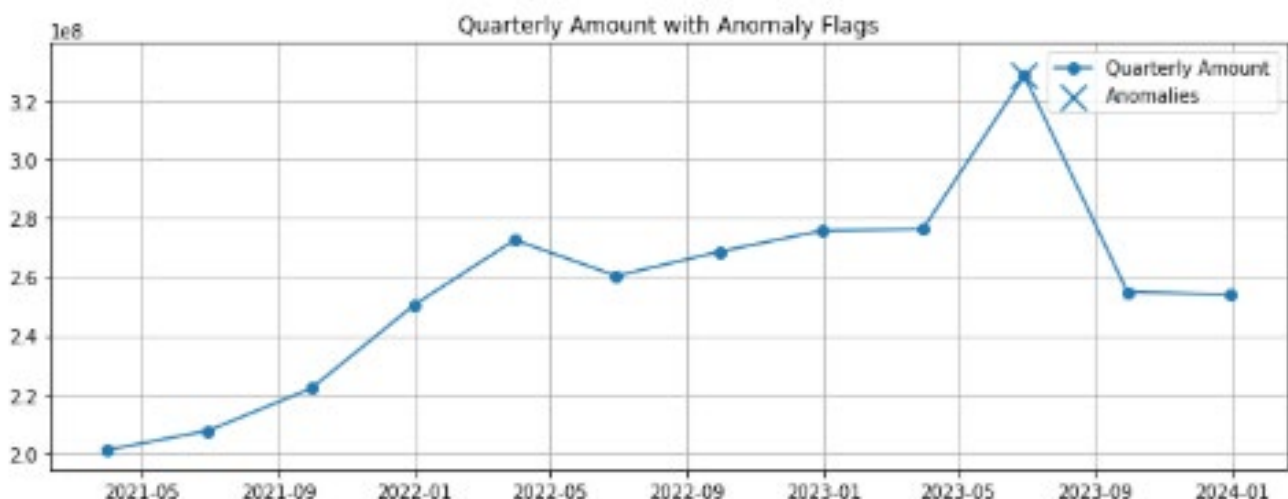
General Payment: Time Series

The visuals shown runs time-based visuals and modeling on the payment_date within the general_payments_record_id_level_clean.csv. It provides counts and amounts by year, by nature of payment, by quarter. It shows anomalies by quarter and anomalies in billing spikes specified by rolling quarterly amounts for individual PIs as well.

The Monthly Payment Amount by Year shows the monthly trends across each year; surprisingly there are no major deviations across each year. The monthly plot also shows that there is a quarterly basis to the payments made; however, it does not follow a typical fiscal quarter pattern, since the peaks are on the 2,5,8, and 11 months. The survey collection method that produces this data does not follow this pattern either, since it is collected annually. It also doesn't follow the typical federal quarters or corporate quarters.



Based on the Monthly Payment Amount by Year, the next step was to create a quarterly time series.

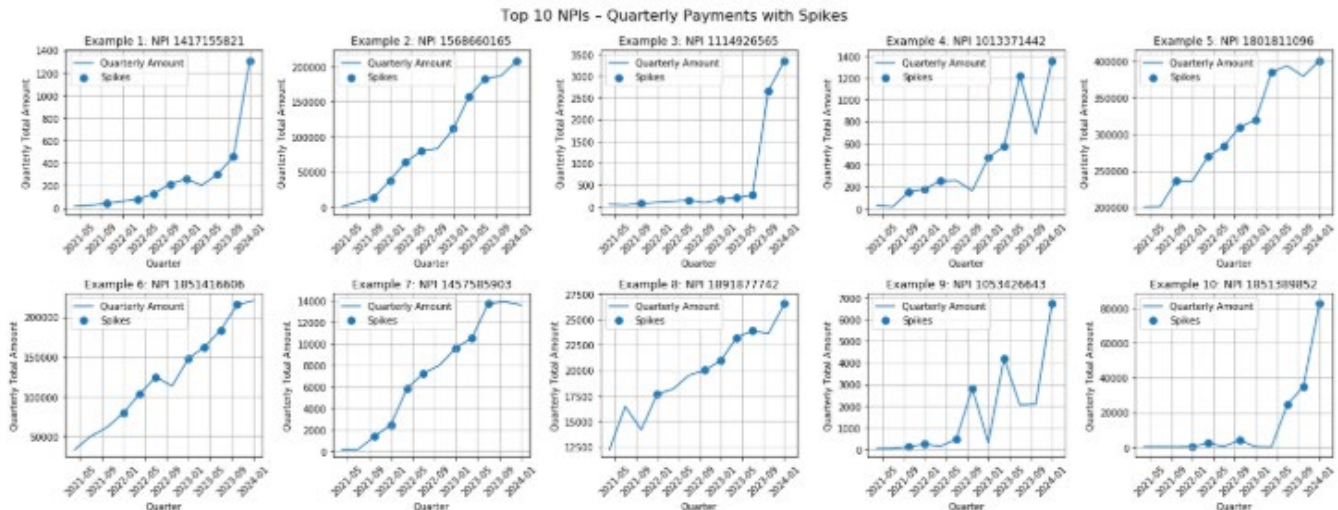


There is one standout quarter in mid-2023 where the manufacturing payment amount rises sharply, marked as an anomaly, and the very next quarter drops back down near the prior baseline and stays flat into late 2023 and early 2024. A one-year window was used to compute a centered rolling mean and rolling standard deviation. For each quarter, a Z score was calculated as the deviation of the observed from the rolling mean, scaled by the rolling standard deviation; quarters were then flagged when the absolute Z score exceeded 1. Because the series was not transformed, this anomaly is being judged on raw levels in a non-stationary, trend-influenced series.

The Augmented Dickey-Fuller test results indicate that the quarterly time series is likely non-stationary.

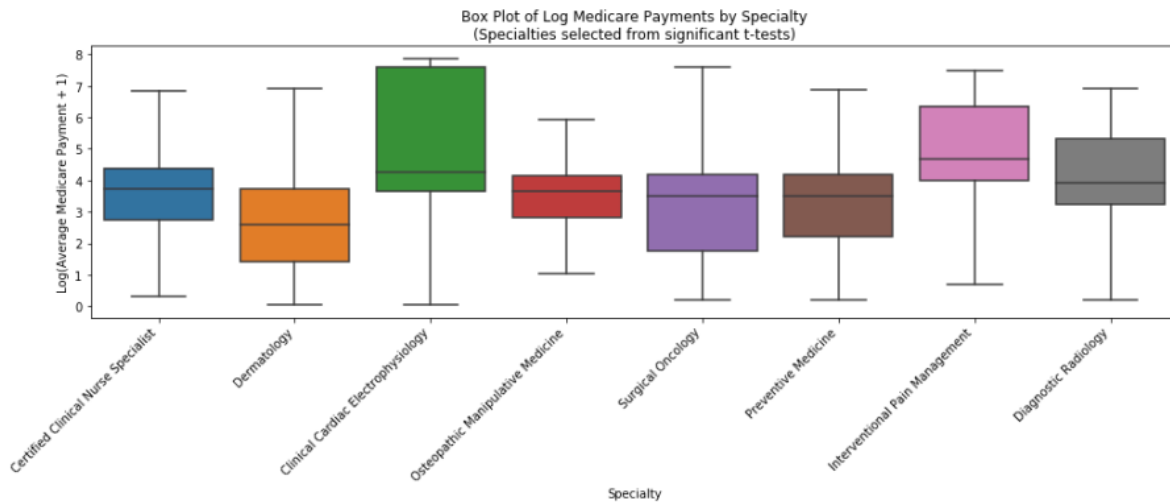
- ADF Statistic: -2.34
- p-value: 0.157

The series does not revolve around a stable average. In practice, this means the data may need transformation before downstream modeling or anomaly detection.

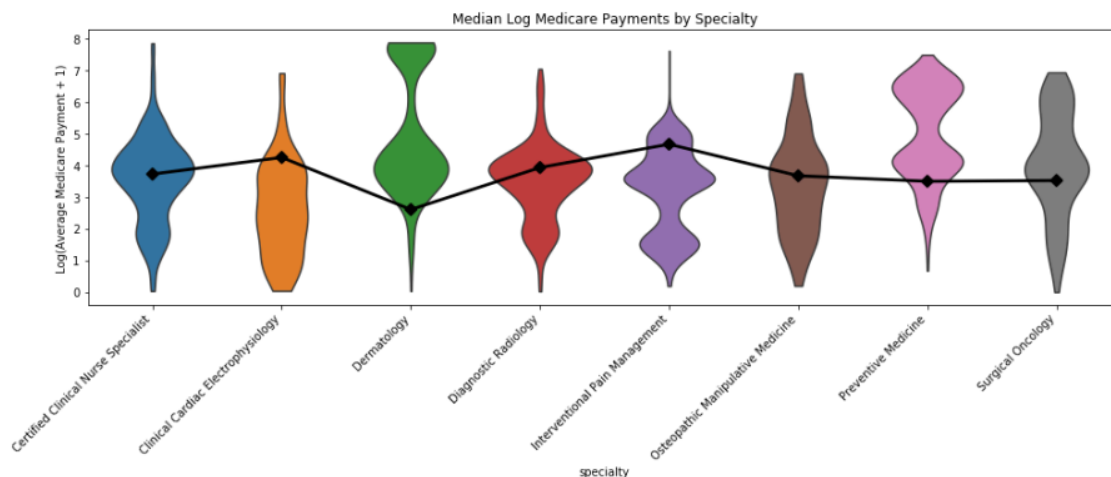


The line series based on the rolling quarterly payments by provider were created to visually show any billing spikes, a specified area of interest from our SMEs. The provider's payments were aggregated first into quarter-by-quarter totals and then the dollar amounts were plotted as a time series over a rolling quarterly baseline. Billing spikes were chosen by comparing each quarter's total to the provider's recent rolling history and flagging quarters that exceeded the expected range by a large standardized deviation, so the marker reflects an unusually sharp jump relative to that provider's own prior pattern rather than a high dollar amount on its own.

Specialties



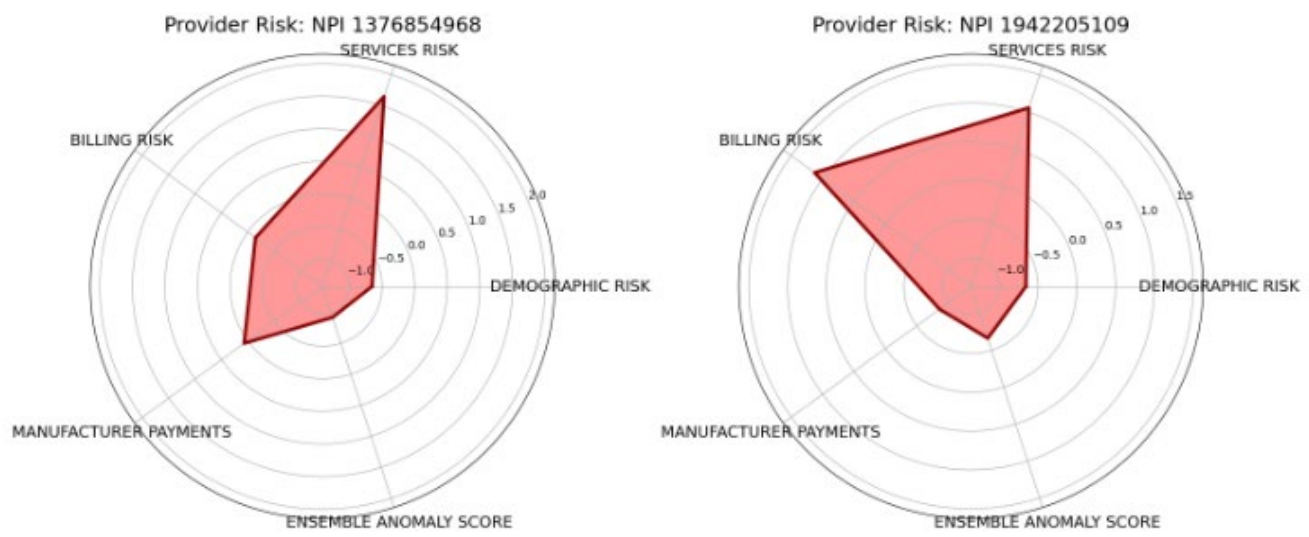
The box plots are included to summarize the median, interquartile range, and overall spread of log-transformed Medicare payments. Outliers are suppressed to prevent extreme values from obscuring central patterns.



While violin plots emphasize distributional shape, the box plots highlight differences in median payment levels and dispersion, making cross-specialty comparisons more interpretable.

Provider Risk Score Radar Chart

The radar chart summarizes provider risk by converting each domain into a standardized z-score that measures how far a provider deviates from the mean pattern observed within the available data. For each provider, relevant features are aggregated into five conceptual domains: demographic risk, services risk, billing risk, manufacturer payment exposure, and ensemble anomaly score. Each domain’s raw values are averaged and transformed into z-scores, allowing the model to display how unusually high or low a provider sits relative to peers. Providers with consistently positive deviations exhibit higher-than-expected risk signals across multiple dimensions, while negative or near-zero values indicate domain stability or alignment with normal patterns.



Although the z-score framework is an easier method and a useful way to visualize the radar charts in the example case study project, the current research plan outlines an ensemble of unsupervised models that directly contribute anomaly scores rather than relying on randomized anomaly scores. The current provider risk score radar has both the mean-normalized deviations and a temporary anomaly score, but the future versions of the radar chart could change into a predetermined multi-feature anomaly profile from the machine learning framework or incorporating additional knowledge from SMEs as outlined in the table below.

RISK FACTOR	LOW RISK	MEDIUM RISK	HIGH RISK
DEMOGRAPHIC RISK	Provider in county with low poverty Low senior population Low veteran density	Provider in county with moderate poverty Moderate senior population Moderate veteran density	Provider in county with high poverty High senior population High veteran density
SERVICES RISK	Primarily provides Low-risk services High-risk service percentage < 10% of total services	Primarily provides low to moderate - risk services High-risk service percentage around 10–30% of total services	Primarily provides moderate to high - risk services High-risk service percentage >30% of total services

RISK FACTOR	LOW RISK	MEDIUM RISK	HIGH RISK
BILLING RISK	Stable billing, no spikes	Moderate YOY changes, high amounts	Extreme billing spikes, unstable volume, payments drop off
MANUFACTURER PAYMENTS	Few or no payments	Recurrent small payments	Large/recurring payments, ownership
ENSEMBLE ANOMALY SCORE	Consistent with peers	Mixed anomaly signals	Strong anomalies across multiple models

Statistical and Machine Learning Model Planning, Building, and Testing

Research Plan

This section is the current research plan of our healthcare fraud project. The project plan has gone through multiple iterations as we have read additional papers and met with subject matter experts.

Analytical Flowchart

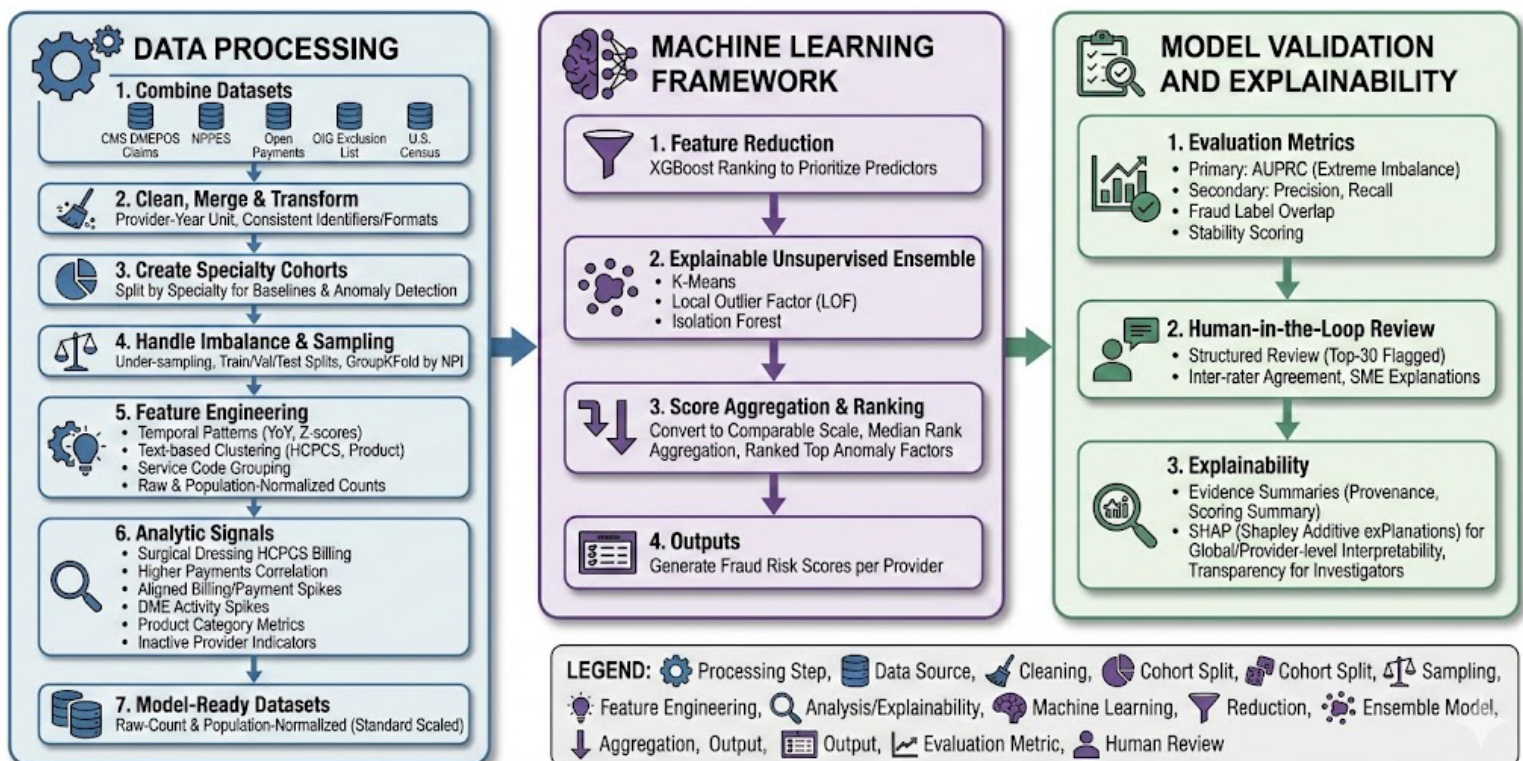


Image generated using Google's Nano Banana.

Data Processing

- Combine CMS DMEPOS Claims, NPPES, Open Payments, OIG Exclusion List, and U.S. Census datasets.
 - o Clean, merge, and transform data to a consistent provider-year analytical unit.
 - o Ensure consistent identifiers, formats, and variable definitions across sources.
- Create specialty-specific cohorts
 - o Split data into specialty subsets for more comparable peer baselines and anomaly detection.

- Handle imbalance and sampling design
 - Use under-sampling strategies for modeling datasets.
 - Create Train/Validation/Test splits using random subsampling by class ratio.
 - GroupKFold by NPI (no provider appears in both train/test)
- Feature engineering
 - Temporal patterns
 - Year-over-year (YoY) changes
 - Z-scores for billing/payment spikes
 - Text-based clustering features (HCPCS short description, product category name)
 - Cluster/group based on service codes (e.g., HCPCS) and procedure descriptions
 - Use both raw counts and population-normalized versions (using Census data where relevant)
- Analytics
 - Billing of higher surgical dressing HCPCS codes relative to others in the same specialty
 - Higher payments correlation to higher claims billing
 - Billing spikes aligned with payments
 - Billing spikes for DME activity
 - Product category metrics
 - Inactive providers indicators
- Datasets
 - Maintain two model-ready datasets (e.g., raw-count and population-normalized), both scaled via standard scaling.
 - Have the unified combined dataset but also test on individual domain datasets.

Machine Learning Framework

- Feature reduction
 - Apply feature ranking using XGBoost to reduce dimensionality and prioritize informative predictors.

- Explainable unsupervised ensemble for anomaly detection
 - Train unsupervised models to detect clustering/outliers:
 - K-Means
 - Local Outlier Factor (LOF)
 - Isolation Forest
 - Convert each model output to a comparable scale
 - Rank aggregation (median rank)
 - Produce ranked top anomaly factors to support investigator interpretation.
- Output
 - Generate risk scores for each provider

Model Validation and Explainability

- Evaluation metrics
 - Primary: AUPRC (Area Under the Precision–Recall Curve) for extreme class imbalance
 - Secondary: Precision and Recall
 - Identified fraud label overlap (Given the few fraud labels we have, identify the overlap between inliers and outliers)
 - Stability scoring (determining how well the models can provide the same outputs)
 - Structured human-in-the-loop review of top 30 flagged providers
 - Inter rater agreement or SME explanations
- Explainability
 - Evidence summaries: Include data provenance and summary of scoring
 - Implement SHAP (SHapley Additive exPlanations) to:
 - Interpret feature contributions at the global and provider-level
 - Improve transparency and usability of risk scores for fraud investigators

Result evaluation is more challenging in this project because unsupervised, explainable machine learning is inherently difficult to validate, particularly with respect to quantifying false positives. In the beginning, we planned to rely exclusively on supervised learning approaches since that is the primary methodology used in fraud detection and other similar works. However, as the project evolved, we explored positive-unlabeled (PU) learning as an alternative, but this direction was not supported by our stakeholders or industry subject-matter experts. Industry experts primarily deploy unsupervised models, so we have chosen that direction. We also considered incorporating a geospatial component at one stage, yet stakeholders ultimately did not view this perspective as sufficiently useful for the project's objectives.

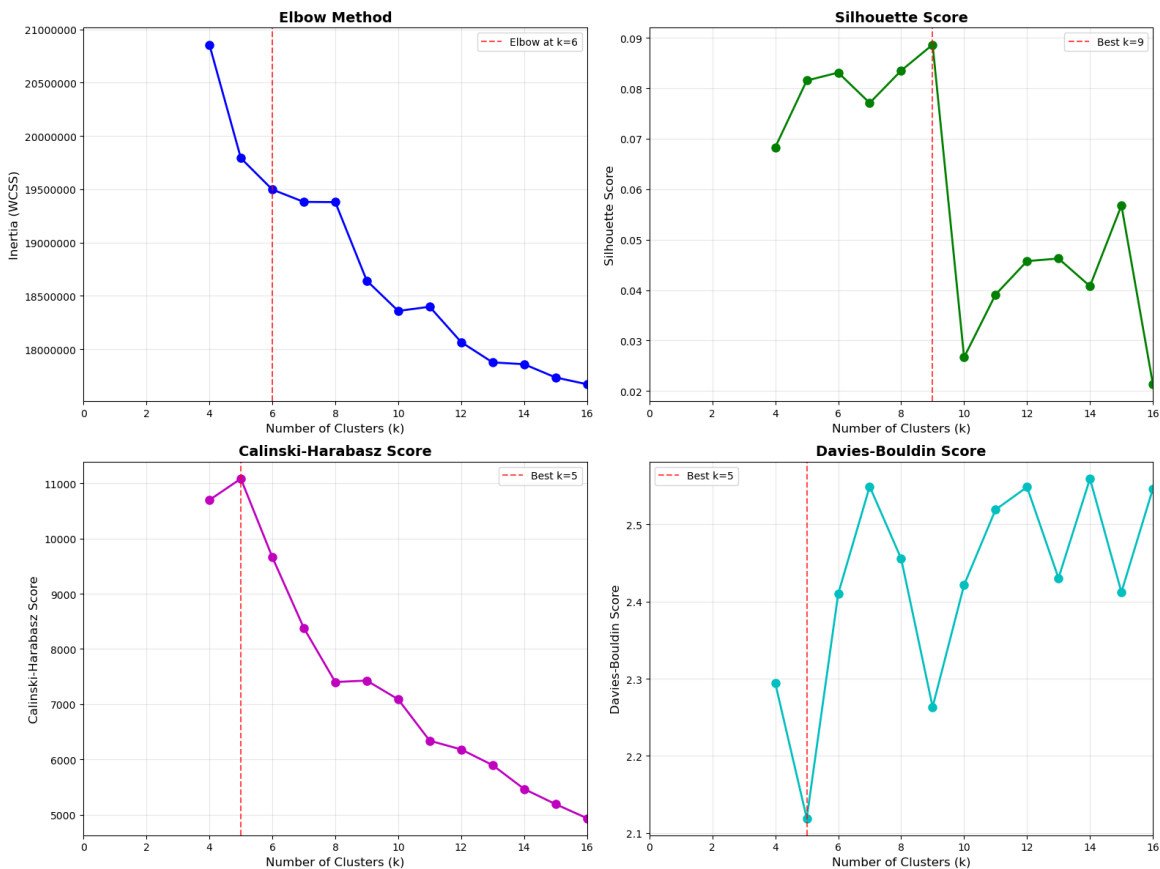
For an initial test of supervised machine learning, we attempted to implement the extreme gradient boosting algorithm for decision trees in a similar manner to *Johnson & Khoshgoftaar (2023)*. We modified to an 80%-20% train-test split compared to their five-fold cross validation and applied the same hyperparameters to the model. This model failed to predict any fraud in testing. One benefit of applying this technique was that we received feature importance scores in the form of information gain. By reiterating on a subset of important features we may be able to recover a useful model. Furthermore, we will consider additional techniques such as oversampling and under sampling.

Statistical and Machine Learning Model Validation, Visualization, and Interpretation

Final Model Selection

At this stage, our work is not yet at the point of final model selection; however, we have run baseline supervised and unsupervised approaches to ensure our modeling strategy is aligned with the project’s core aim of identifying credible fraud risk signals that can support oversight and prioritization. These initial runs are therefore best interpreted as directional evidence that helps validate feature relevance, interpretability, and operational usefulness

K-Means

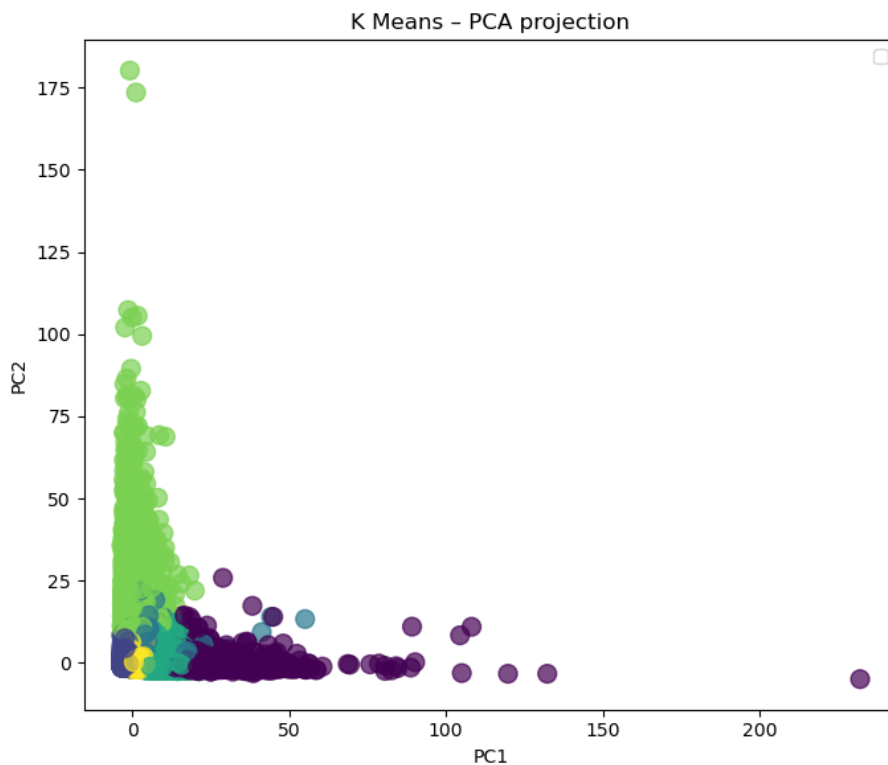


Based on a dataset size of 193,015 samples, the best parameters for a general K-Means clustering algorithm is shown here. Using the Calinski-Harabasz Score, Davies-Bouldin score, elbow method, and the silhouette score the optimal number of clusters was manually defined at 6 since it had the best visual indicators across each plot. It also provides the top features that differentiate clusters.

Best K Means Parameters:

- K= 6
- Inertia= 19,496,901
- Silhouette = 0.083
- Calinski = 9,662
- Davies=2.410

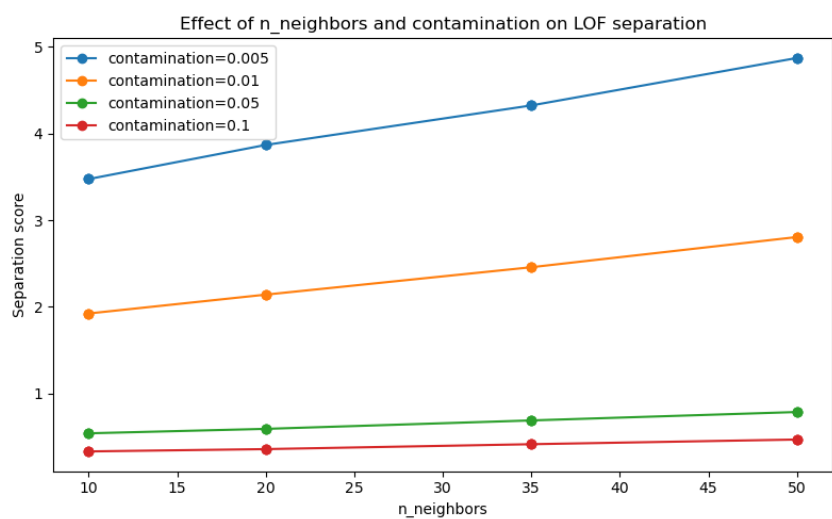
The 2D projection shown via PCA is shown below.



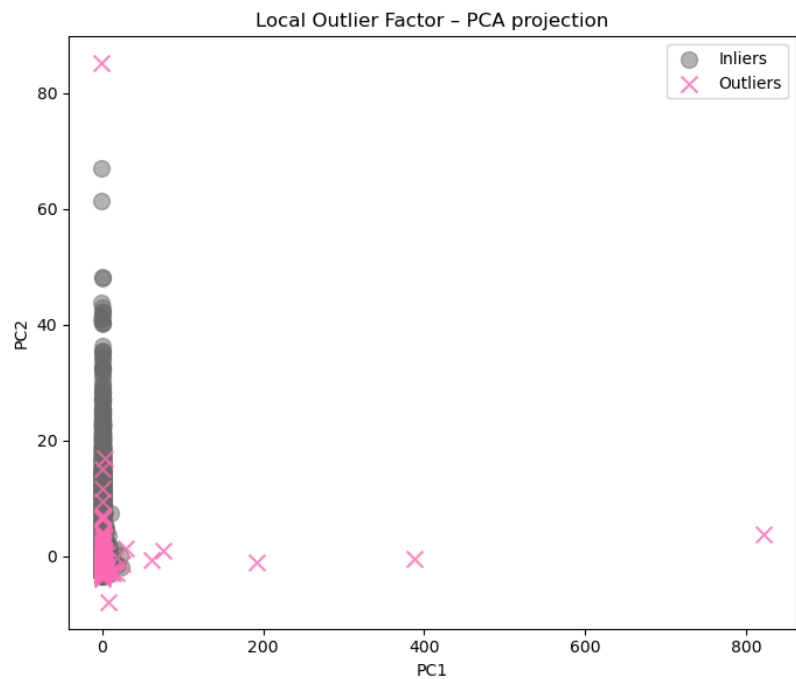
Local Outlier Factor

Local Outlier Factor (LOF) is a common anomaly detection algorithm used on large data. LOF is a density-based method for detecting objects that are unusual relative to their local neighborhood rather than across the global dataset. LOF assigns each object a continuous score by comparing the object's local density to the average local density of its nearest neighbors. Objects with substantially lower local density than their neighbors receive a high LOF score and are flagged as local outliers. Entities that operate in significantly sparser regions, i.e., provider billing patterns that are unusual among their peers receive high LOF scores and are prioritized as potential fraud cases.

The LOF was created using scaled data and the labels from the fraud exclusions. Via Grid Search the best parameters were: (contamination': 0.005, 'leaf_size': 10, 'metric': 'minkowski', 'n_neighbors': 50, Separation: 4.87232592281622, 'n_outliers': 120).



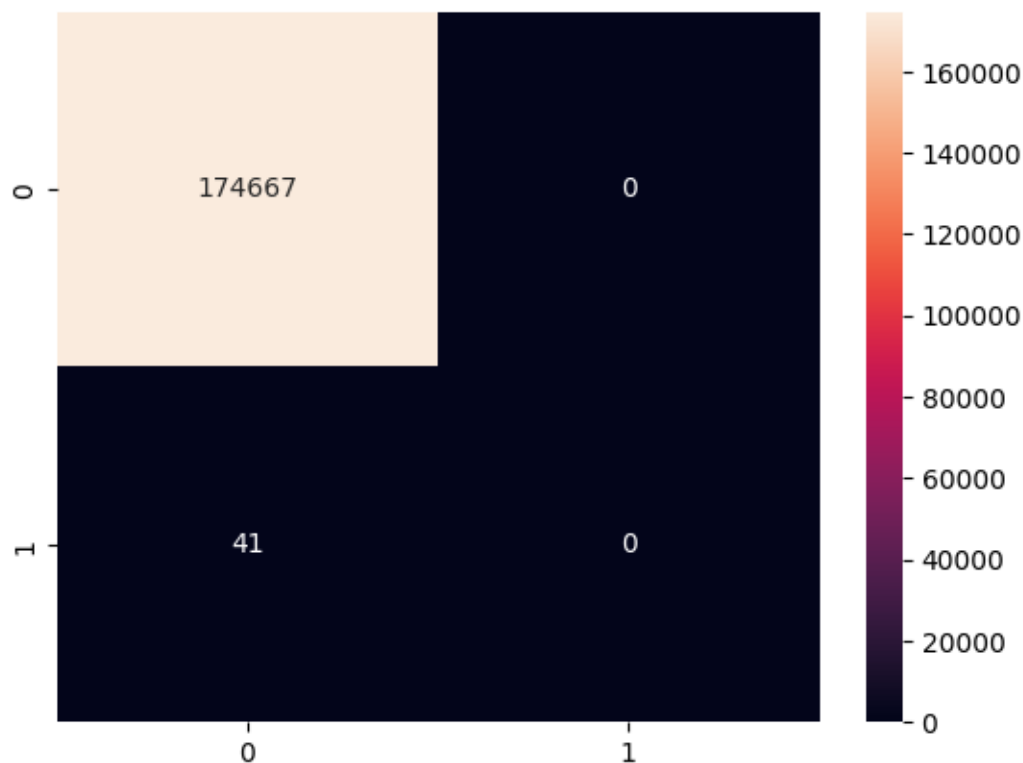
The lowest rate of contamination makes sense in this scenario given the drastic class imbalance. This rate also led to more class separability via the separation score. In addition, most of the data from the unified dataset was considered non anomalous, with only a few points considered outliers. The 2D projection shown via PCA is shown below. Based on this projection, most of the provider data is concentrated along PC1. There are only a few outliers as shown in pink that may warrant review. Additional visualizations and data processing should occur to give the most interpretable anomaly detection for our fraud control stakeholders.



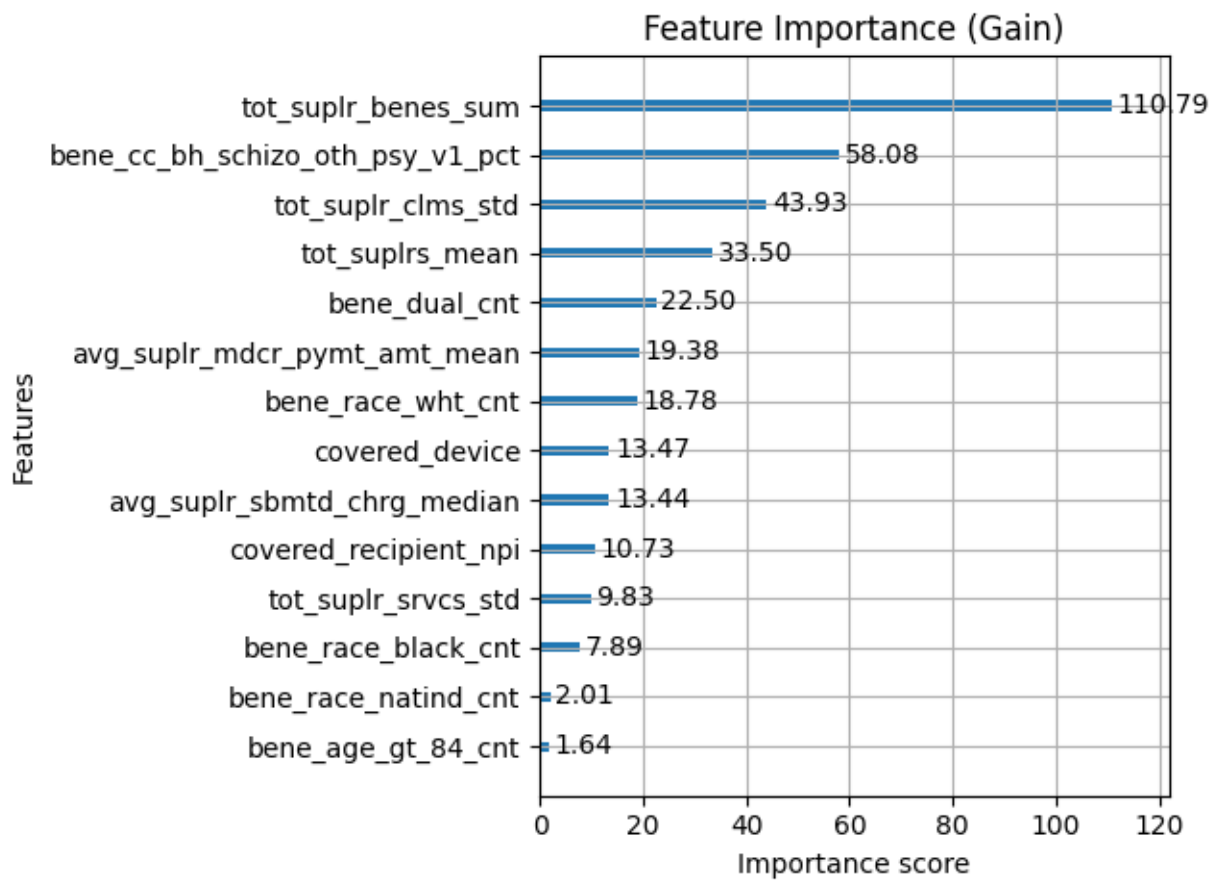
XGBoost

We attempted extreme gradient boosting (XGBoost) on the unified dataset removing features that were known to be uninformative, like provider NPI, or that could present a data leakage problem, like number of all time exclusions. We split the data in a stratified manner, with an 80% - 20% train-test split. We trained the model with default hyperparameters except for the `max_depth` paramter, which was set to 4.

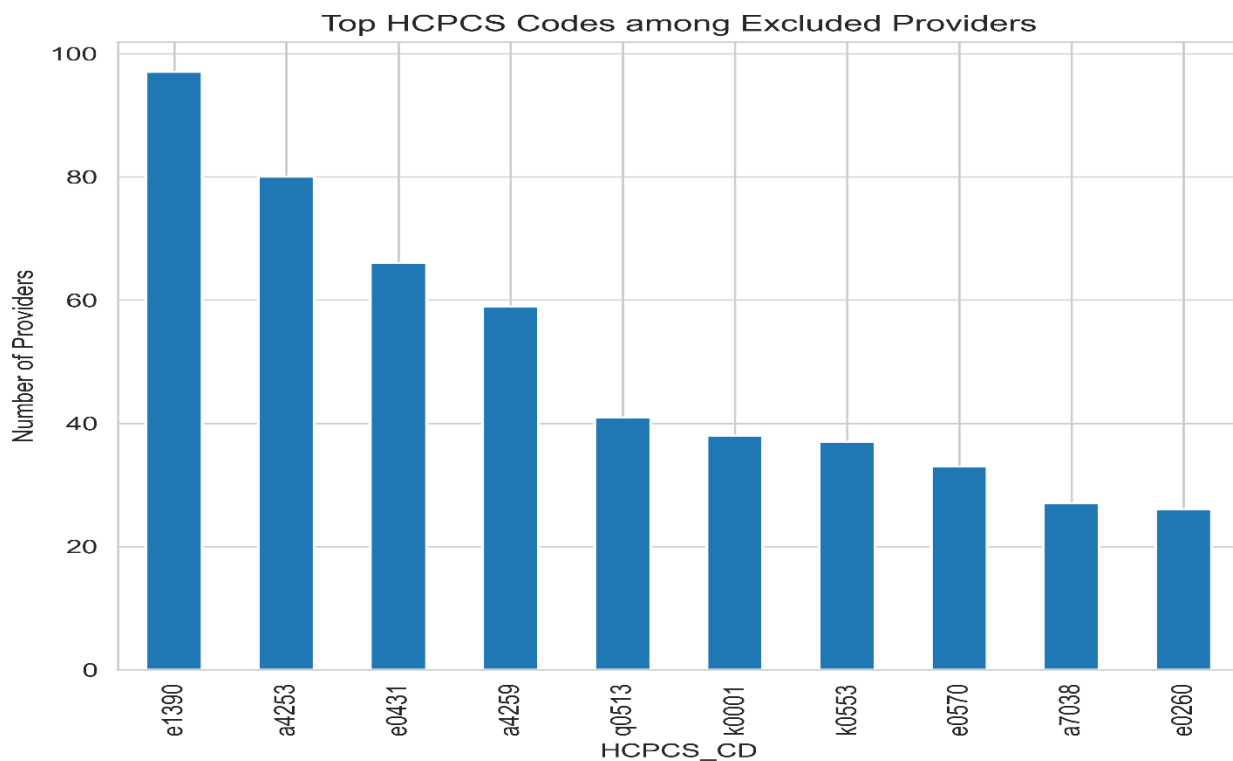
We found the resulting model to be useless upon validation. As the confusion matrix shows, the model predicts no fraudulent providers despite the presence of 41 in the test set. Considering a modified objective function to implement with the model is a potential line of future work.



We extract feature importance from the model in terms of information gain. By far, the most important feature by this metric is the sum of total supplier beneficiaries of a referring provider for a program year. Understanding why this feature contributes so significantly is a matter of future work. XGBoost is a supervised learning method, and we hope a pivot into unsupervised methods will be more fruitful and shed light onto the mechanisms driving the initial feature ranking.



Data Conclusions



As discussed in the EDA and visualization section, this chart highlights the HCPCS codes most frequently associated with excluded referring providers, many of which are related to oxygen supply. Given that the average age of Medicare beneficiaries is around 60, these types of services are commonly needed, which explains why claims for these codes appear frequently.

- e1390 claims : 368,324 (The most provided service)

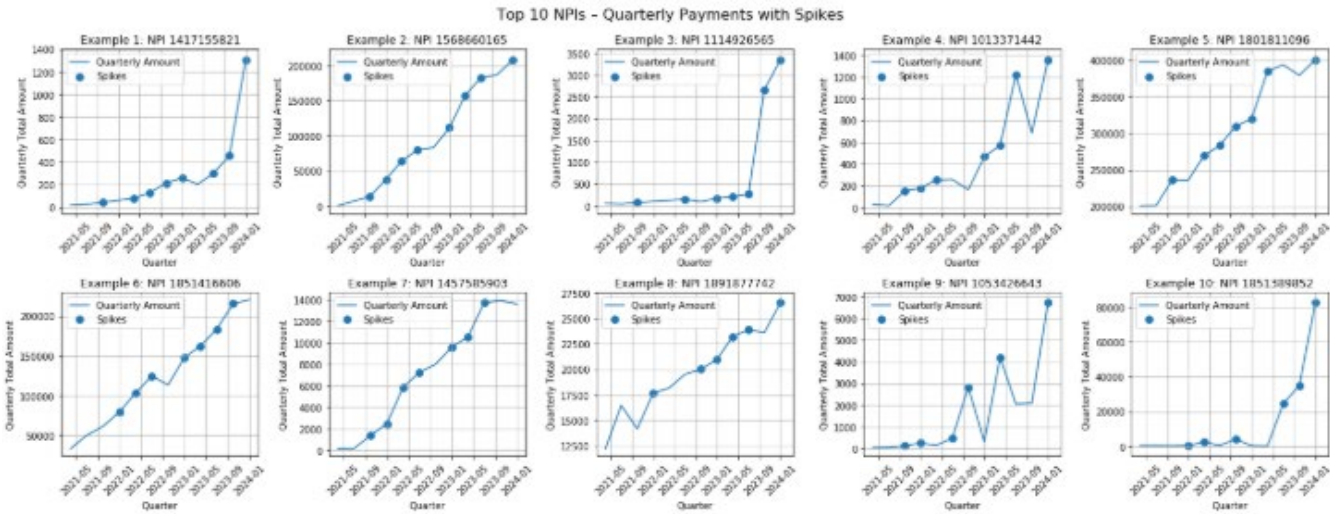
Therefore, a chi-square test was conducted to determine whether the top 10 HCPCS codes were significantly associated with provider exclusion, using a significance level of 0.05.

Results:

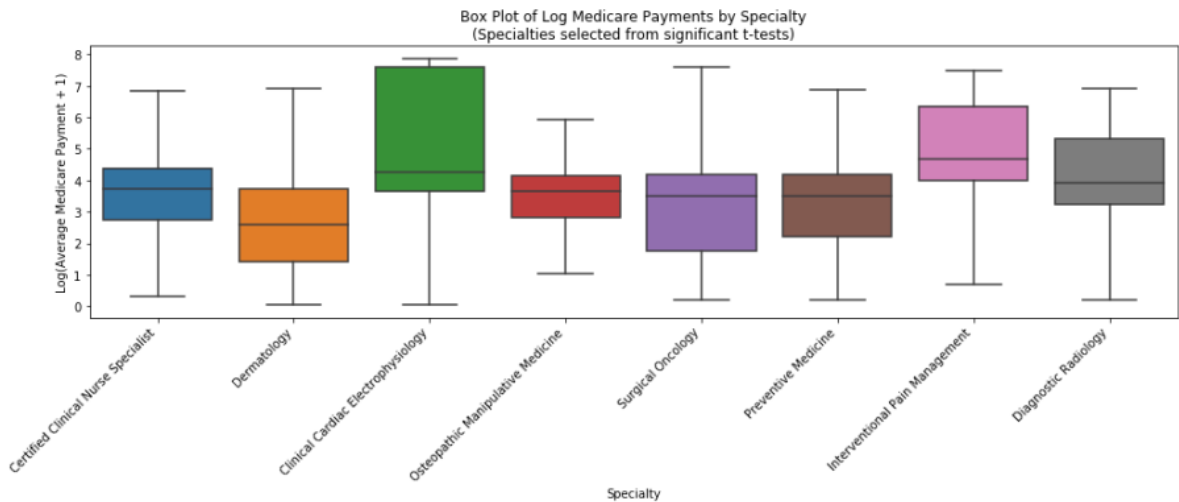
- Chi-square: 25.32
- p-value: 0.00264

Since the p-value is lesser than 0.05, we can reject the null hypothesis and conclude that service codes are significantly associated with provider exclusion. This also suggests that HCPCS code can play an important role

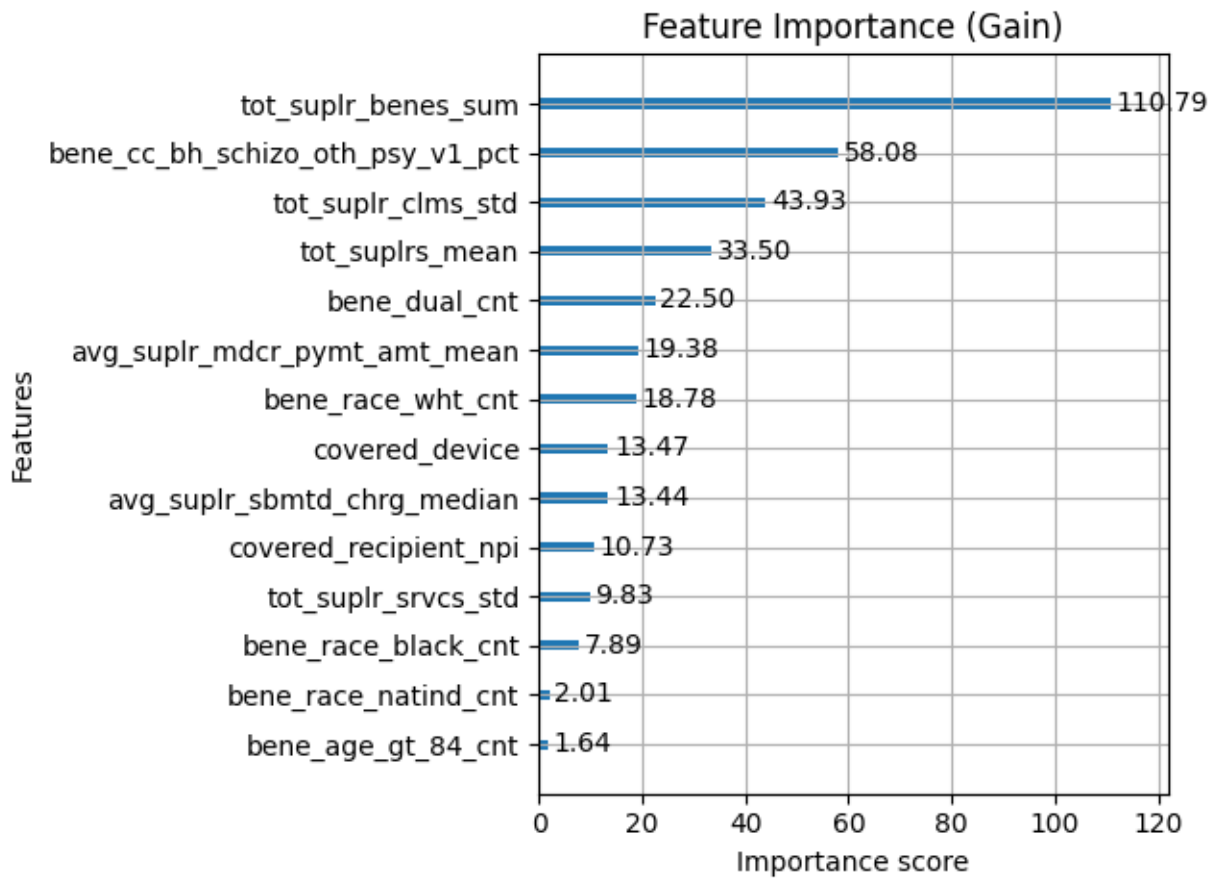
in fraud analysis.



The line series based on the rolling quarterly payments by provider were created to visually show any billing spikes, a specified area of interest from our SMEs. The provider’s payments were aggregated first into quarter-by-quarter totals and then the dollar amounts were plotted as a time series over a rolling quarterly baseline. Billing spikes were chosen by comparing each quarter’s total to the provider’s recent rolling history and flagging quarters that exceeded the expected range by a large standardized deviation, so the marker reflects an unusually sharp jump relative to that provider’s own prior pattern rather than a high dollar amount on its own.



The box plots are included to summarize the median, interquartile range, and overall spread of log-transformed Medicare payments. Outliers are suppressed to prevent extreme values from obscuring central patterns. T-tests were performed to identify the specialties where Medicare payments were significantly different.



We extract feature importance from the model in terms of information gain. By far, the most important feature by this metric is the sum of total supplier beneficiaries of a referring provider for a program year. Understanding why this feature contributes so significantly is a matter of future work. XGBoost is a supervised learning method, and we hope a pivot into unsupervised methods will be more fruitful and shed light onto the mechanisms driving the initial feature ranking.

Data Story / Product Development and Report Generation

Data Story

Background

Our project “Detecting Fraud Signals in Centers for Medicaid and Medicare Services (CMS) Data using Machine Learning on Class Imbalanced Big Data” focuses on the healthcare fraud, waste and abuse that drains billions of taxpayer dollars each year and undermine the integrity of federally funded programs. Bad actors are motivated by opportunity, rationalization, and financial pressure to use healthcare as a means for fraudulent activity.

One particularly vulnerable area involves Durable Medical Equipment, Prosthetics, Orthotics, and Supplies (DMEPOS) which includes items such as wheelchairs, prosthetics, and medical supplies. DMEPOS is a frequent target of fraud and has a high improper payment rate. In Fiscal Year (FY) 2024, CMS identified \$1.9 billion (21.4%) in improper payments related to DMEPOS. CMS is taking significant actions to crush fraud conducted by DMEPOS suppliers. The traditional fraud detection

system takes time and relies heavily on manual audits and retrospective reviews, which are often slow, resource-intensive, and reactive rather preventive.



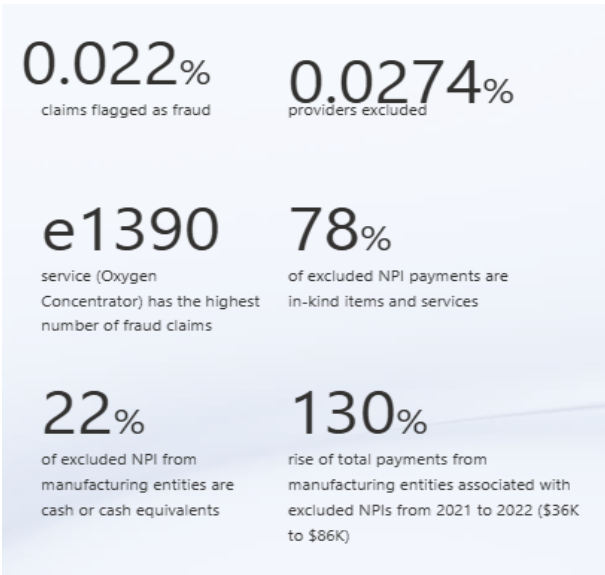
Fig 1. Fraud triangle. Source: [Fraud Conference News](#).

To assist with this problem, our approach...

...uses an analytical framework that integrates CMS DMEPOS claims, NPPES, Open Payments, the OIG Exclusion List, and U.S. Census indicators into a standardized provider-year dataset. We apply an explainable unsupervised ensemble to model baseline “normal” behavior within each specialty, then flag providers who are statistically atypical without relying on large numbers of confirmed fraud labels. Key signals include year-over-year changes, z-scored billing and payment spikes, and utilization mix patterns derived from HCPCS codes and product categories. Each provider receives a risk score by scaling and aggregating anomaly outputs across models (K-Means, LOF, Isolation Forest), producing a ranked list for triage. For transparency, the score is paired with top contributing anomaly factors and an evidence summary that documents the specific behaviors driving the flag, enabling faster, more consistent investigative prioritization.

Fraud in America

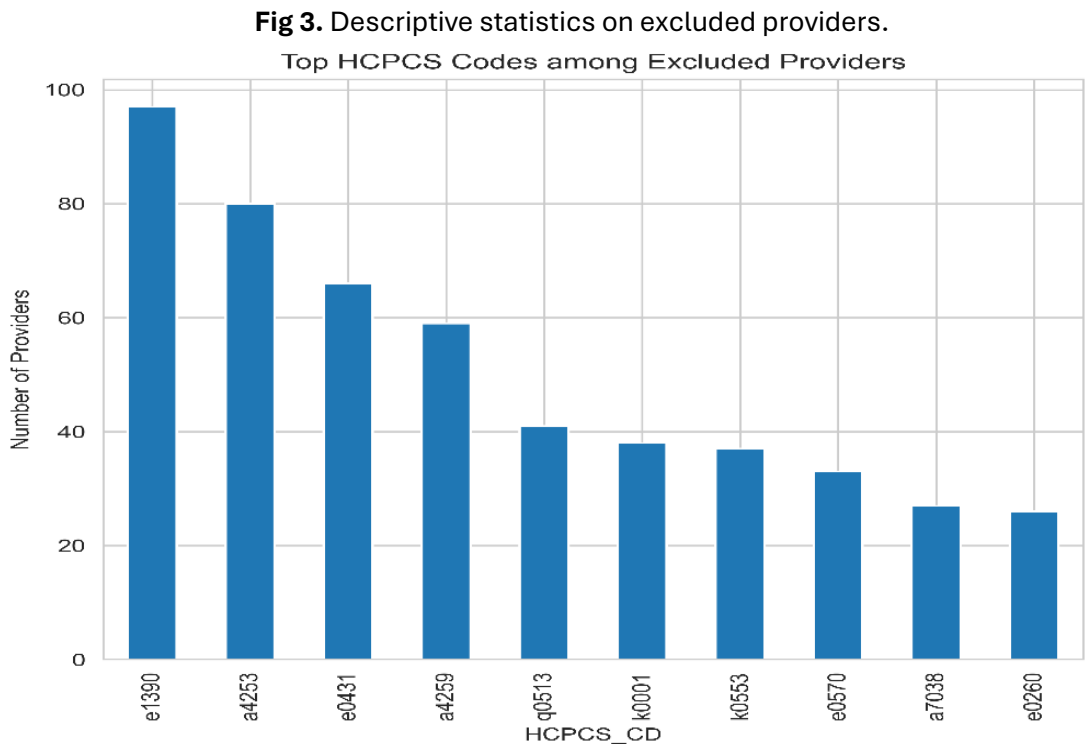
Some key findings from our data (2021-2023) are shown here.



While the statistics show that only a tiny portion of claims and providers are affected by fraud, it allows us to measure the scale and even assess the risk of the problem, highlighting the services, providers, or payment types that are most vulnerable, enabling targeted oversight.

Without these figures, it would be quite easy to underestimate the financial and patient-care impact of fraud, even if it is statistically rare, and miss patterns that could prevent and control loss.

Fig 2. Descriptive statistics on excluded providers. Primary DMEPOS Items Targeted by Excluded Providers



This visualization highlights the HCPCS codes most frequently associated with excluded referring providers. Most of these services are related to oxygen supply equipment, which is often billed monthly as rentals. This creates opportunities for fraud, either through upcoding or by failing to deliver the billed equipment.

Provider Specialties

An analysis of Medicare payment distributions shows that certain provider specialties such as interventional pain management, diagnostic radiology, and clinical cardiac electrophysiology exhibit higher median payment levels and wider variability compared to others. This pattern indicates that elevated billing volume is not uniformly distributed across specialties and may mask atypical billing behavior within high-intensity service areas.

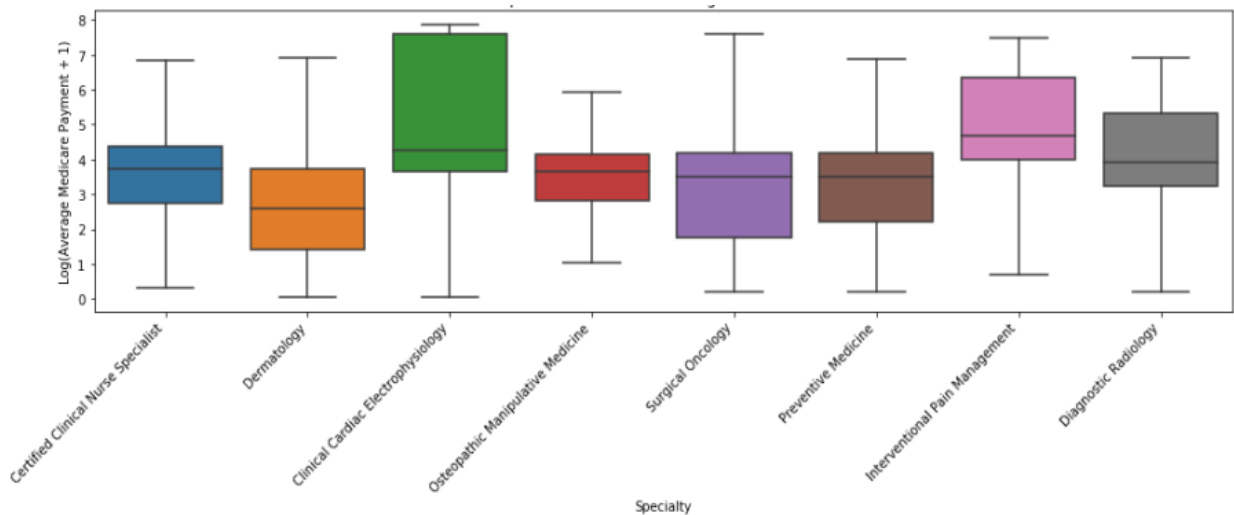
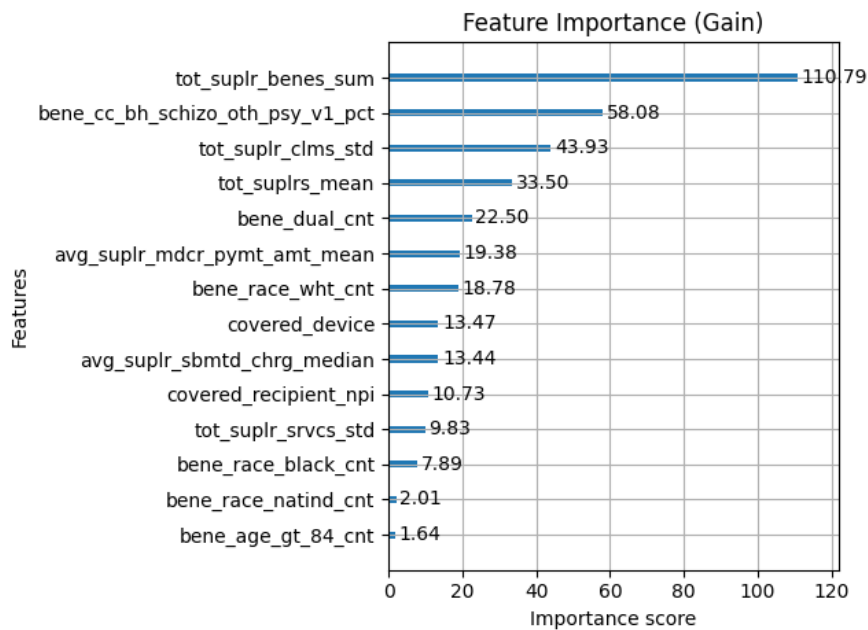


Fig 5. Boxplots by provider specialty.

Following Characteristics Indicative of Excluded Providers

Using current machine learning methodology, we used XGBoost to rank characteristics that differentiate between OIG excluded providers and normal providers. By far, the most important feature by this metric

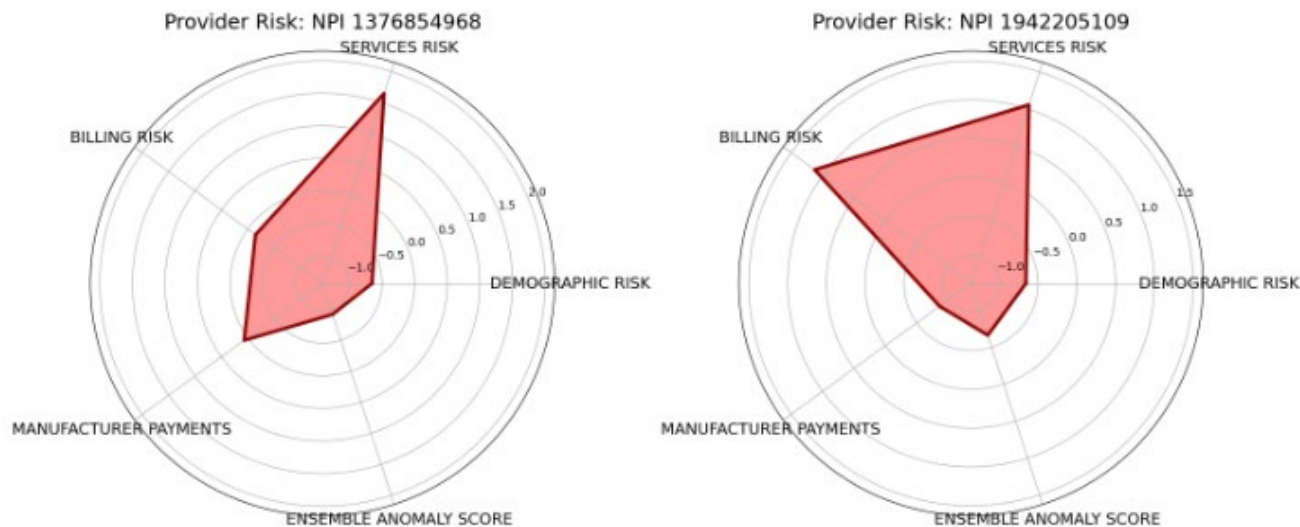


is the sum of total supplier beneficiaries of a referring provider for a program year. Understanding why this feature contributes so significantly is a matter of future work

Fig 6. Feature importances of excluded provider classification.

Different Risk Factors of Fraud

The provider risk score breaks providers into low, medium, and high risk, based on various factors.



Including where they practice, what they bill for, and how they’ve behaved in the past. A supplier in a county with high poverty and a large senior population, offering a high share of “high-risk” services, money flowing in from manufacturers, history of high billing, lands farther to the outskirts of each circle in red.

Fig 4. Example NPI Radar Charts. The radar chart length is determined by the z score of the average values within each risk factor. The data used here is for example demonstrations only.

RISK FACTOR	LOW RISK	MEDIUM RISK	HIGH RISK
GEOGRAPHIC & DEMOGRAPHIC RISK	Provider in county with low poverty Low senior population Low veteran density	Provider in county with moderate poverty Moderate senior population Moderate veteran density	Provider in county with high poverty High senior population High veteran density
SERVICES RISK	Primarily provides Low-risk services High-risk service percentage < 10% of total services	Primarily provides low to moderate - risk services High-risk service percentage around 10–30% of total services	Primarily provides moderate to high - risk services High-risk service percentage >30% of total services
BILLING RISK	Stable billing, no spikes	Moderate YOY changes, high amounts	Extreme billing spikes, unstable volume, payments drop off

MANUFACTURER PAYMENTS	Few or no payments	Recurrent small payments	Large/recurring payments, ownership
ENSEMBLE ANOMALY SCORE	Consistent with peers	Mixed anomaly signals	Strong anomalies across multiple models

Payments Spike Spotlight

These providers show distinct billing spikes, or quarters where payments from manufacturers are significantly higher than their own payment history based on a 4-quarter rolling z-score. In practice, these spikes may reflect true changes in activity (e.g., new product advertisements, seasonal programs, or one-off speaker events) or potential anomalies that warrant follow-up. Providers with multiple spike quarters are especially important to review, since outliers suggest a shift in billing behavior.

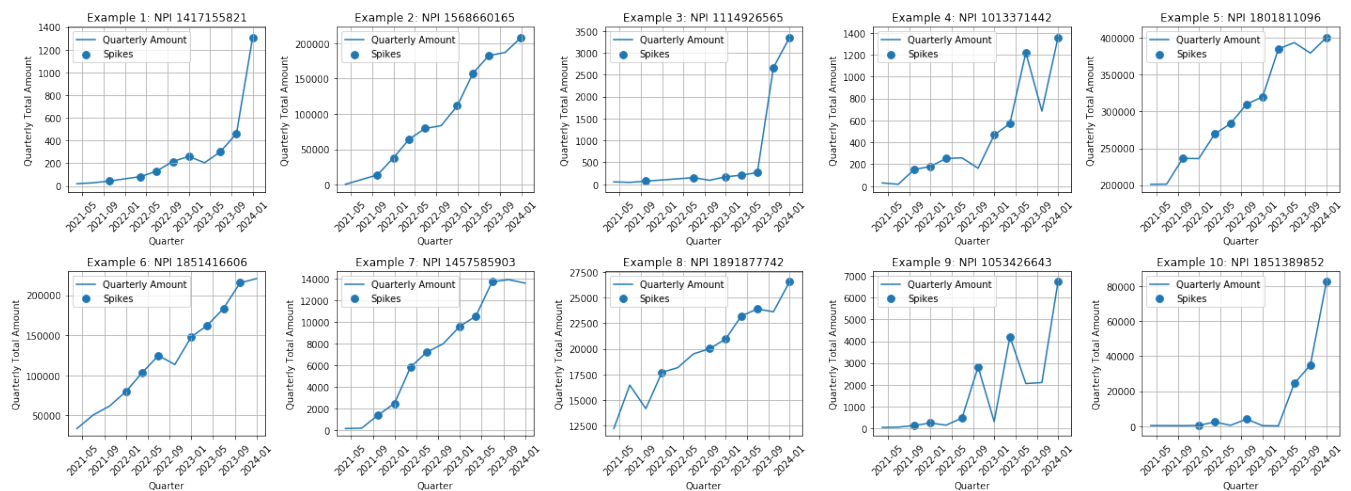


Fig 6. Top 10 NPIs with manufacturing payment spikes.

Conclusion

By combining descriptive analytics, feature ranking, and anomaly detection methods, we developed a framework that enhances the ability of oversight teams to identify suspicious behavior and prioritize cases for review. While this system does not predict fraud, it reveals the structural, financial, and behavioral markers that consistently align with confirmed cases. These insights help lay the groundwork for more efficient, proactive, and data-informed antifraud efforts.

Preventing Healthcare Fraud

If you suspect DMEPOS fraud, contact your provider or supplier or report it at cms.gov/fraud or by calling 1-800-MEDICARE.

Final Information Product

The final intended deliverable for this project is an interactive, investigation-based analytics dashboard designed to assist stakeholders in preventing Medicare and Medicaid fraud, such as the Medicare Fraud Control Unit and Program Integrity. The plan is to present data-driven leads on potential suspicious DMEPOS suppliers and providers via informative visualizations, such as time-series and provider-level trend analyses of payments and billing behavior.

These insights are intended to support investigators in identifying suspicious patterns and enable more informed decision-making during investigation. Additionally, the dashboard will incorporate an AI-powered agent that summarizes relevant external information, such as OIG case articles, healthcare news, and regulatory filings, to provide supporting, case-relevant information and context about flagged providers and suppliers. Together, the dashboard and AI agent function as a decision-support tool that could potentially reduce losses, mitigate systemic risk, ensure compliance, and strengthen oversight.

Reflection

Looking back on our project process so far, if we were to do this again, we would start by defining a more commercializable deliverable that's better suited for production deployment. Knowing upfront that the solution would need to integrate into an existing environment would change both how we design the deliverable and the questions we ask stakeholders. The outcome of a production-ready project likely would have shifted the project's goals and even the stakeholders we chose.

If we had the necessary organizational and bureaucratic support, we would have pursued CMS Limited Data Sets for Researchers (which require a data use agreement and is standard for this sort of project). However, given the constraints of our database and the administrative burden, we were limited to using public data. Another improvement would be to define our research hypotheses earlier. We focused on the aims and deliverables, but we should have created a list of testable hypotheses based in the data we had available.

Why is this Data Science

This project constitutes a data science and analytics driven investigation because it applies systematic, reproducible, and statistically grounded methods to extract actionable fraud risk insights from large, heterogeneous public datasets. Rather than relying on rule-based heuristics or manual review, the work integrates data engineering, exploratory analysis, feature engineering, unsupervised learning, and

model interpretability to address a real-world problem characterized by limited labeled outcomes and high operational complexity.

A core data science contribution of this project is the integration and harmonization of multiple large-scale public datasets including DMEPOS claims, CMS Open Payments (general and ownership payments), the OIG List of Excluded Individuals and Entities (LEIE), Medicare Enrollment, and U.S. Census data across inconsistent schemas, reporting granularities, and temporal coverage. This required advanced data shaping techniques such as entity resolution, specialty-level aggregation, geographic normalization, and temporal alignment. These activities reflect foundational data engineering and data management competencies central to the data science discipline.

The analytical framework further demonstrates data science expertise through feature engineering and representation learning. Financial payment distributions, ownership concentration metrics, temporal utilization indicators, and county-level socioeconomic variables are transformed into interpretable fraud risk signals. The project leverages unsupervised and weakly supervised learning approaches including anomaly detection, principal component analysis, and peer-group benchmarking to identify high-risk patterns without reliance on extensive labeled fraud data. This approach reflects a practical data science response to real-world constraints where ground truth is sparse or delayed.

Model interpretability is a defining component of this work. Rather than prioritizing predictive accuracy alone, the project emphasizes transparent and explainable analytics, producing specialty-level risk profiles with feature attribution that support investigative decision-making. Techniques such as feature importance analysis and structured risk scoring ensure that results are auditable, interpretable, and suitable for operational deployment of key requirements in regulated public-sector environments.

Finally, this project plans to translate analytical outputs into actionable insights for Medicaid integrity units, fraud investigators, and policy analysts which align directly with the applied focus of the Data Science and Analytics discipline. The development of dashboards, risk rankings, and contextualized signals demonstrates the full data science lifecycle: from data acquisition and modeling to communication and decision support. Collectively, this work showcases applied expertise in data engineering, statistical learning, and explainable analytics, firmly establishing it as a rigorous and impactful data science project.

Publication Target and References

Publication Target

The stakeholders generally read articles and newsletters written by the National Association for Medicaid Program Integrity, Office of the Inspector General articles and releases, CMS publications, or other forums that are locked behind security clearance membership. The most realistic journal that we would target this report for is the Springer Journal of Big Data, since most of the relevant studies were published in this journal.

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